

FROM PLANETARY SIGNALS TO TELESCOPE TIME

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10—09—2026





HOW MANY RV MEASUREMENTS DO I NEED?

A proposal needs to answer:

- What signal am I trying to measure?
- Which instrument can measure it?
- What precision can I realistically obtain per exposure?
- How many measurements are required?
- Can I demonstrate that the requested observations achieve the science goal?

SCIENCE REQUIREMENT

What exactly do we want to do?

- Detect planets
- Measure masses
- Resolve orbital periods
- Disentangle dynamical and activity signals

What do we measure?

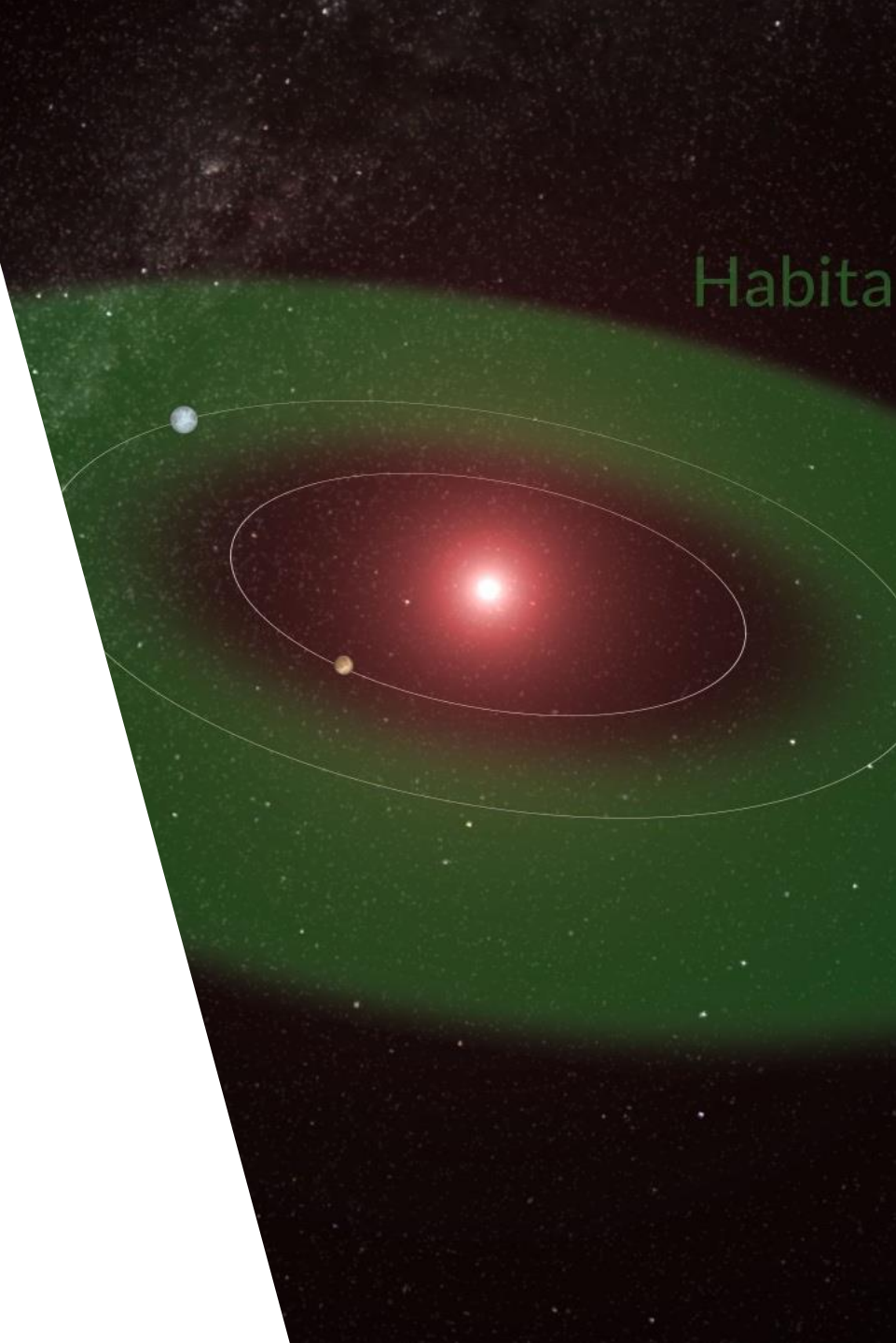
- FAP/FIP/logZ
- Mass precision
- Period uniqueness

Habita

FROM PLANET MASS TO RV SIGNAL

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

$$K \propto M_p P^{-1/3} M_\star^{-2/3}$$



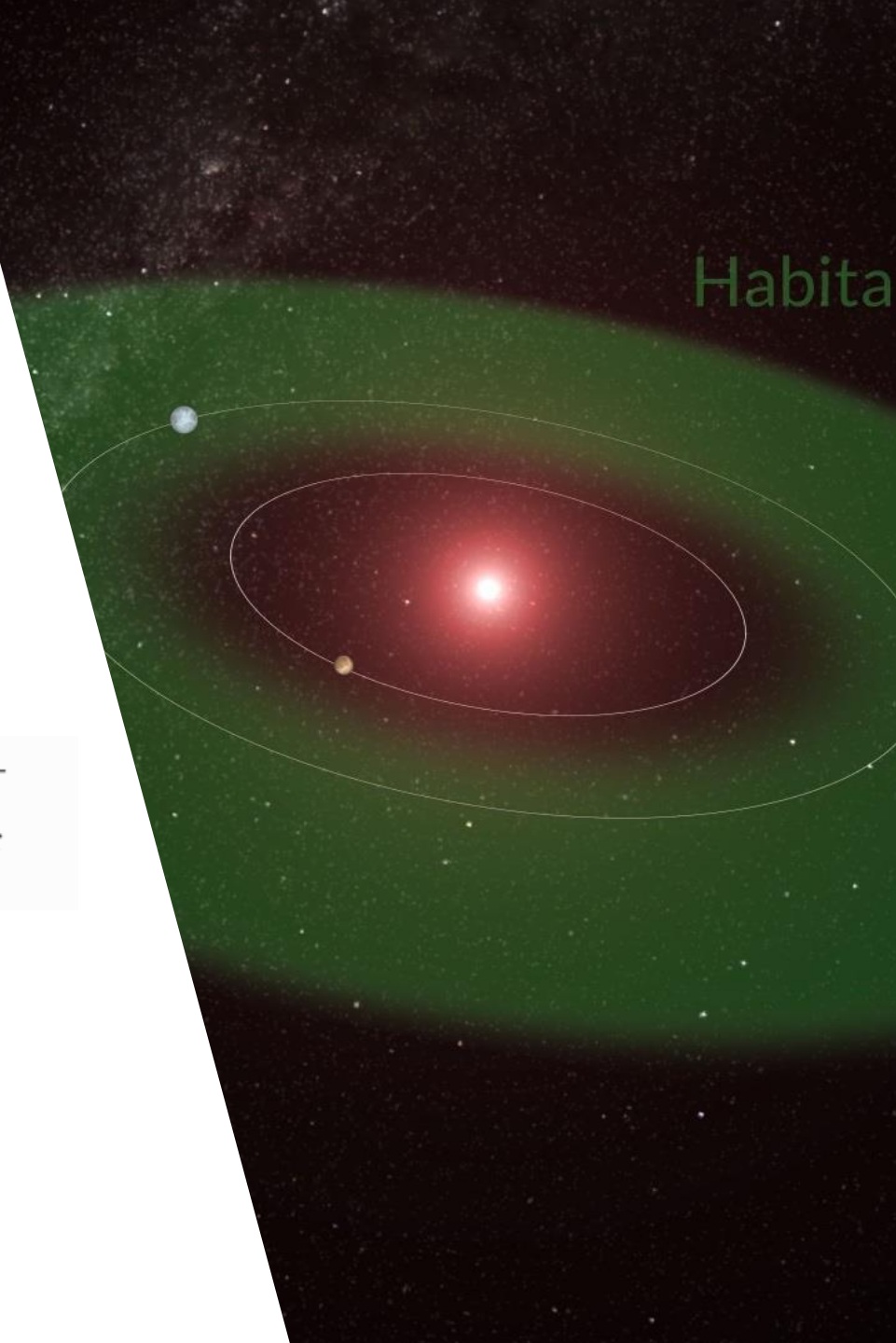
Habita

FIRST ORDER ESTIMATE

$$\sigma_{\text{RV}} = \sqrt{\sigma_{\text{photon}}^2 + \sigma_{\text{instrumental}}^2 + \sigma_{\text{stellar}}^2 + \sigma_{\text{other}}^2}$$

$$\sigma_K \simeq C \frac{\sigma_{\text{RV}}}{\sqrt{N}}.$$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{\text{RV}}}{\sqrt{N}},$$



Habita

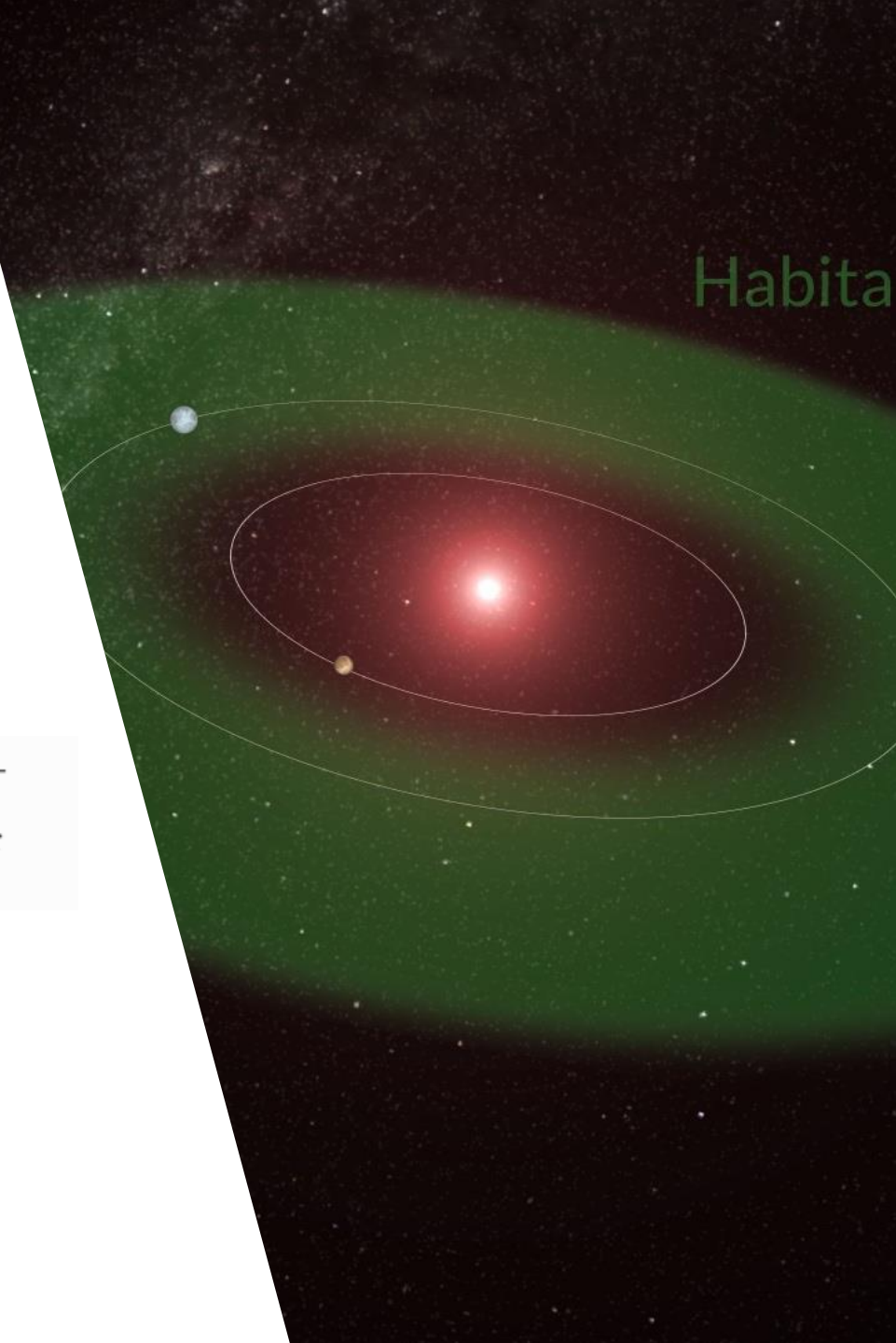
FIRST ORDER ESTIMATE

$$\sigma_{\text{RV}} = \sqrt{\sigma_{\text{photon}}^2 + \sigma_{\text{instrumental}}^2 + \sigma_{\text{stellar}}^2 + \sigma_{\text{other}}^2}$$

ETC
or
Previous data

Instrument floor
(Literature)

Previous data
or
Activity vs RV
relationships



PHOTON NOISE

Target Input Flux Distribution

☒ Template Spectrum	G2V (Kurucz) ▾	Redshift z = 0.00	Target Magnitude and Mag.System: ☒ Vega ☐ AB <i>Magnitudes are given per arcsec² for extended sources</i>
☐ MARCS Stellar Model	Teff=4000 log(g)=-0.5 [Fe/H]= 0 M= 1 ▾	V ▾ = 10	
☐ Upload Spectrum	Select...		
☐ Blackbody	Temperature: K		

Instrument Setup

Telescope feed: 1 UT ▾

Resolution/detector mode: singleHR21 ⓘ *Single UT, high spectral/RV resolution, 2x1 binning and slow readout*

Exposure Time: 900 s

- **Radial velocity precision:**
 - Wavelength: **551.236 nm** (order 111 center)
 - Exposure time: **900 s**
 - S/N: **129.203**
 - RV precision: **0.46 m/s**



ETC “MISFIRES”

Target Input Flux Distribution

GJ 1002 (M5, V13.84)

☒ Template Spectrum	M2V (Pickles) ▾	Redshift z = 0.00	Target Magnitude and Mag.System: V ▾ = 13.84 ☒ Vega ☐ AB <i>Magnitudes are given per arcsec² for extended sources</i>
☐ MARCS Stellar Model	Teff=4000 log(g)=-0.5 [Fe/H]= 0 M= 1 ▾		
☐ Upload Spectrum	Select...		
☐ Blackbody	Temperature: K		

Instrument Setup

Telescope feed: 1 UT ▾

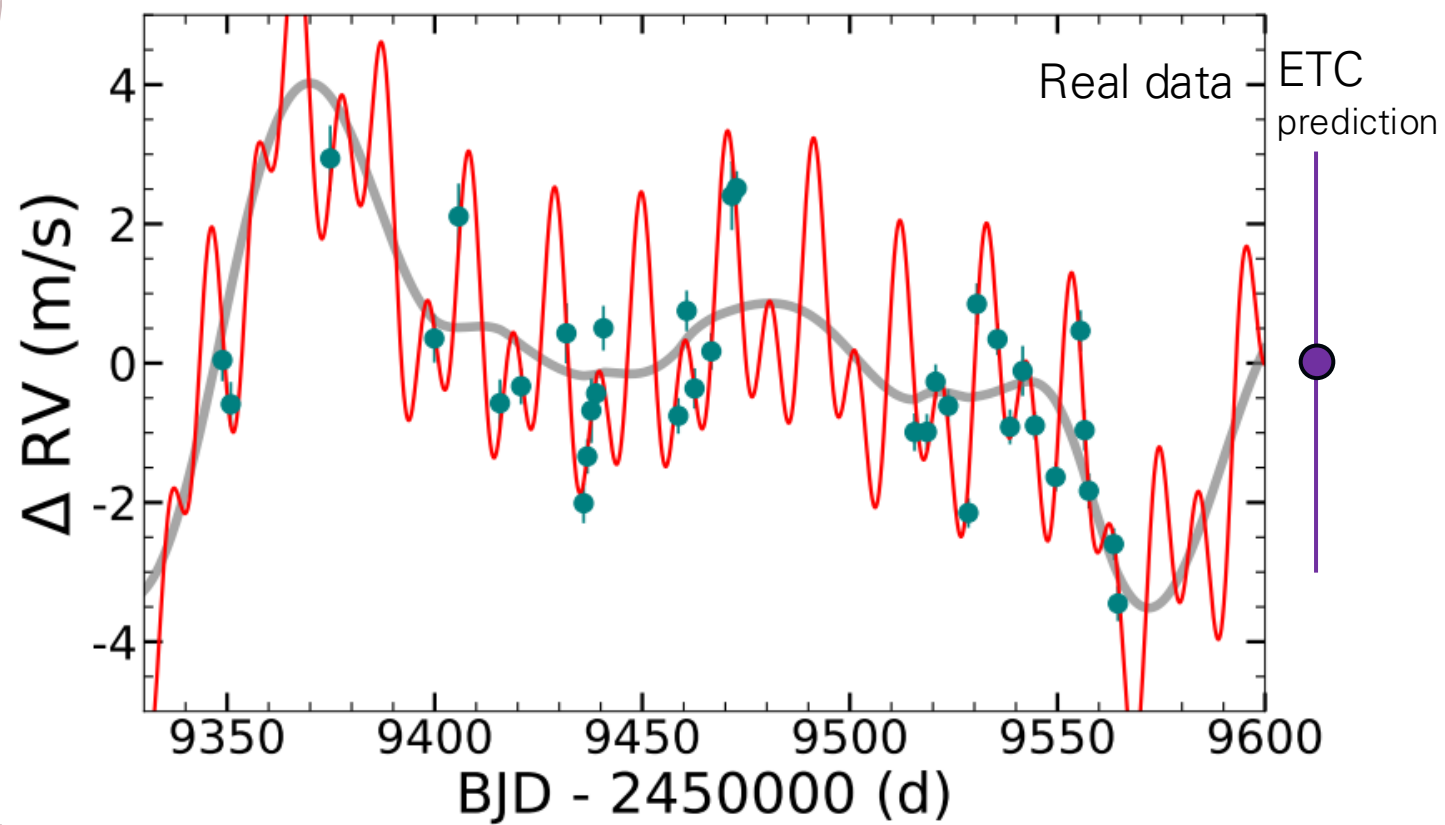
Resolution/detector mode: singleHR21 ⓘ *Single UT, high spectral/RV resolution, 2x1 binning and slow readout*

Exposure Time: 900 s

- **Radial velocity precision:**
 - Wavelength: **551.236 nm** (order 111 center)
 - Exposure time: **900 s**
 - S/N: **20.7259**
 - RV precision: **2.89 m/s**



ETC “MISFIRES”



Suárez Mascareño et al. 2023



INSTRUMENT PRECISION

ESPRESSO (VIS) 10 cm/s

EXPRESS (VIS) 10 cm/s

MAROON-X (VIS) 30 cm/s

NEID (VIS) 30 cm/s

HARPS (VIS) 0.8 m/s

HARPS-N (VIS) 0.8 m/s

NIRPS (NIR) 1 m/s

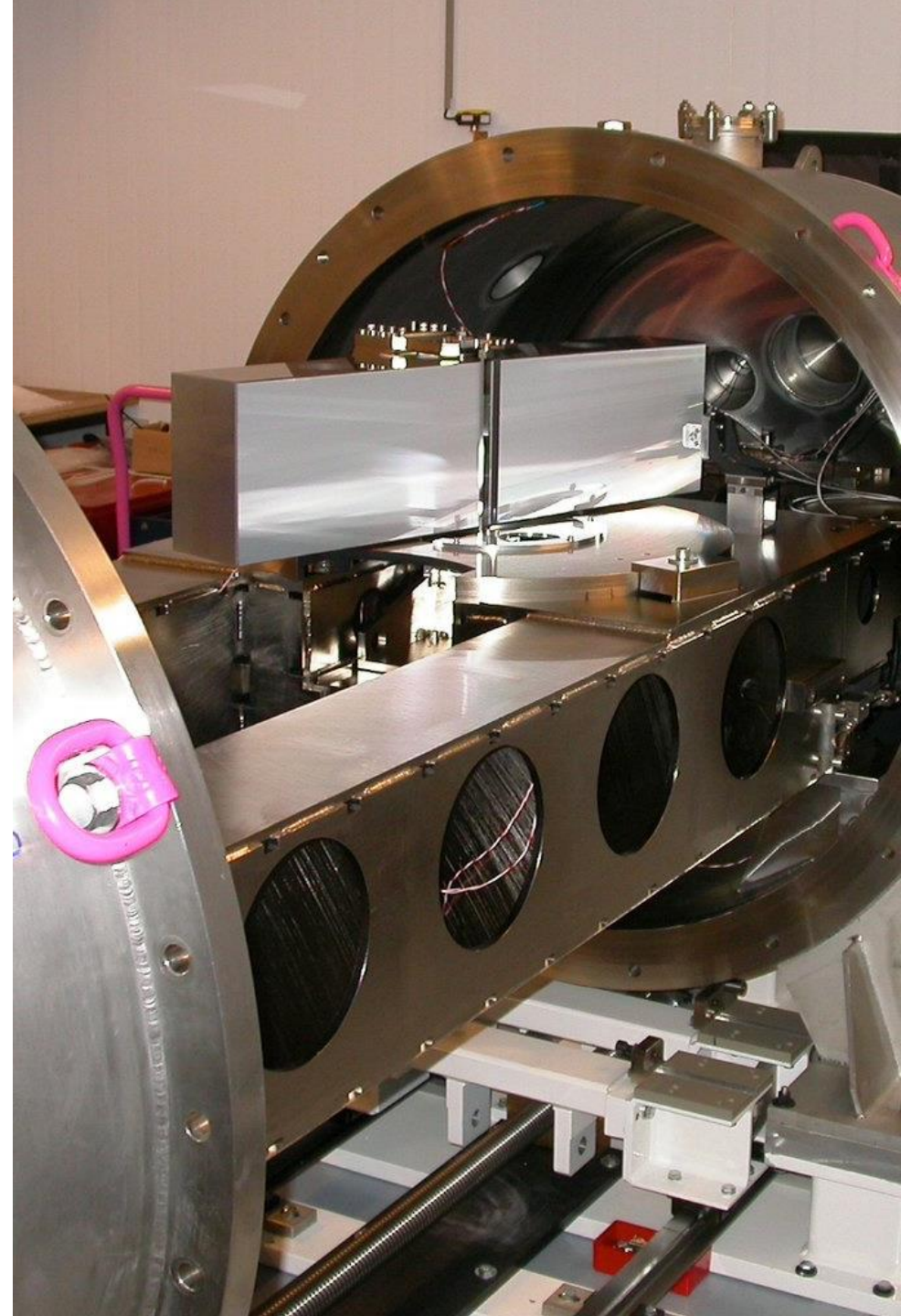
PFS (VIS) 1 m/s

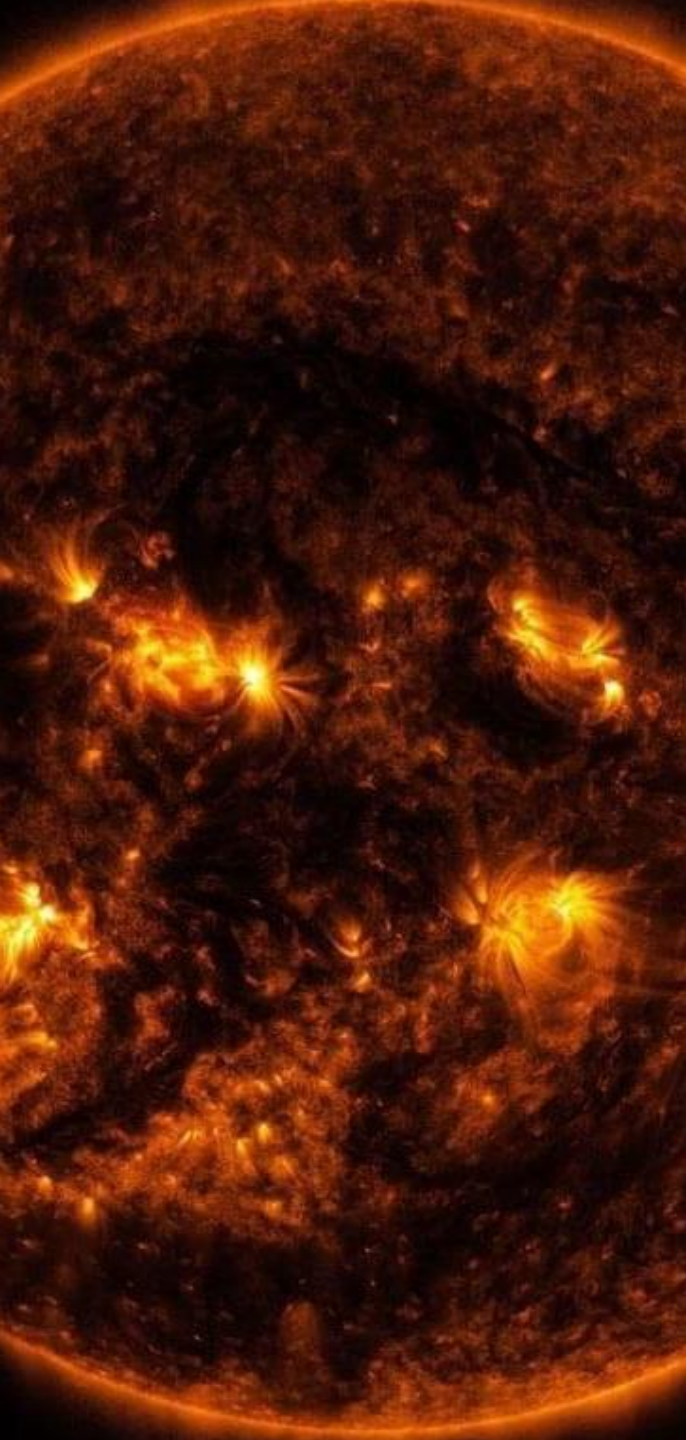
CARMENES (VIS) 1 m/s

HIRES (VIS) 1-2 m/s

SOPHIE (VIS) 2 m/s

SPIROU (NIR) 2 m/s



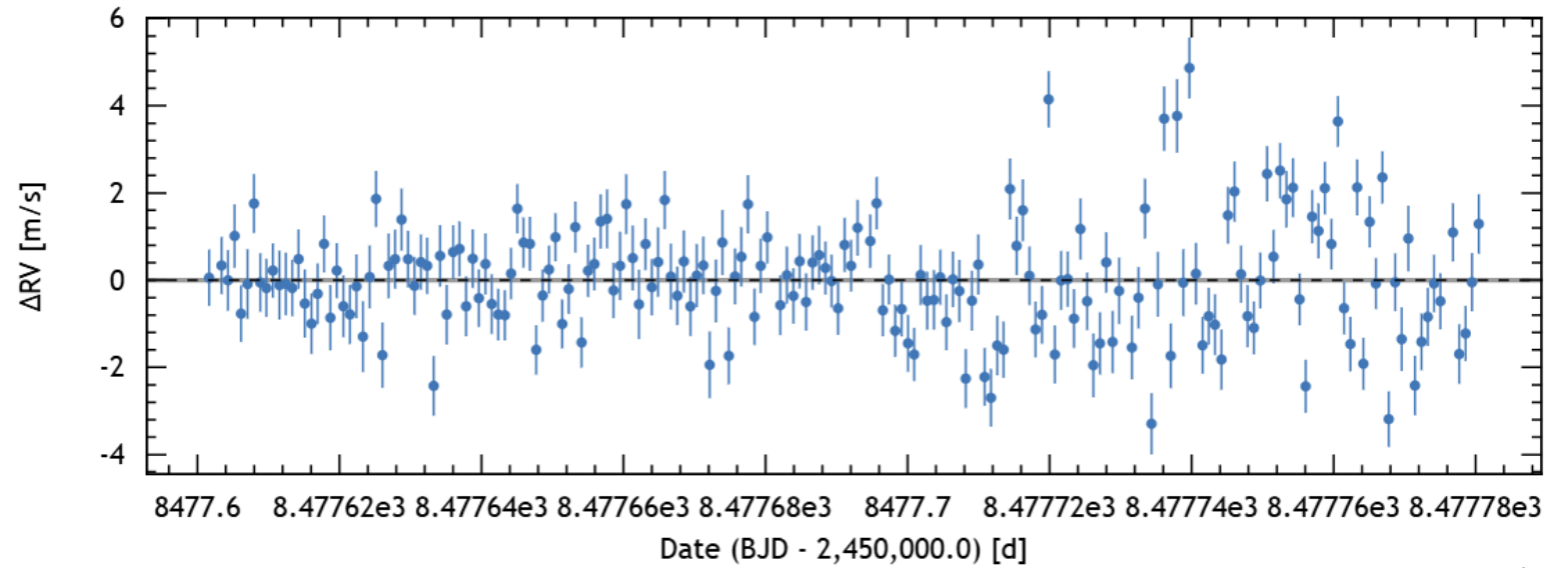


STELLAR ACTIVITY

- Oscillations (P-modes): Minutes
< 2 m/s
- Granulation/Supergranulation: 1-2 hours to 1-2 days
< 2 m/s
- Stellar rotation: < 1 day to > 100 days
< 1 m/s to > 1 km/s
- Stellar cycles: Years
< 1 m/s to > 10 m/s

STELLAR ACTIVITY

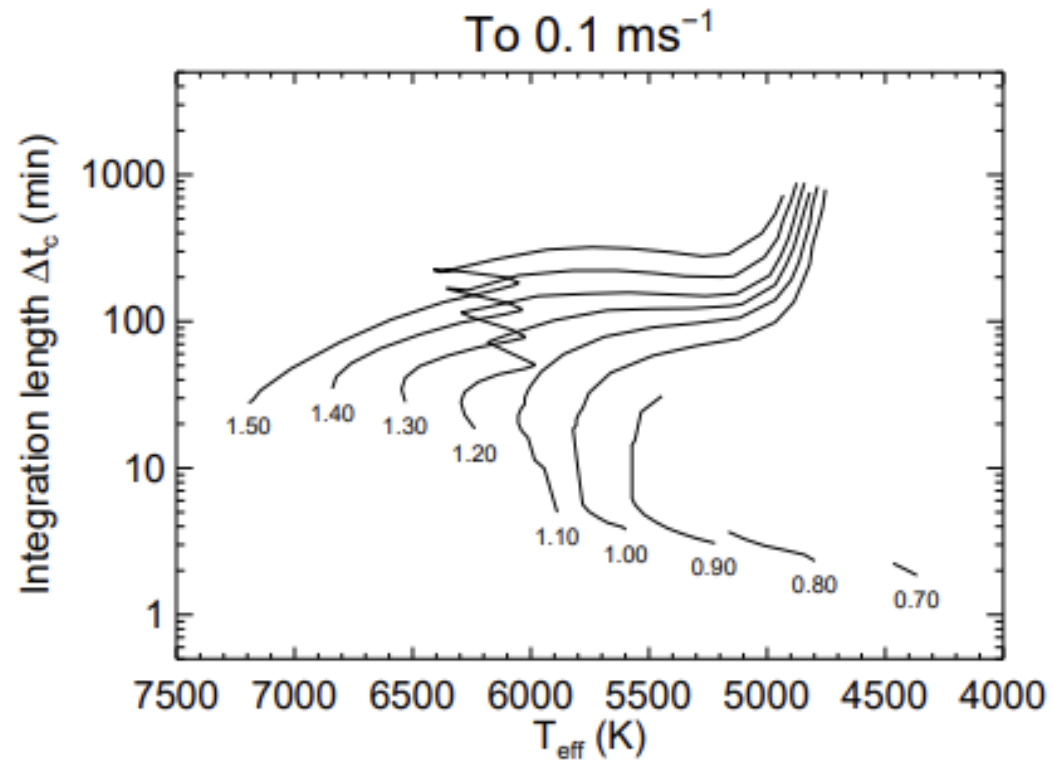
- Oscillations (P-modes): Minutes

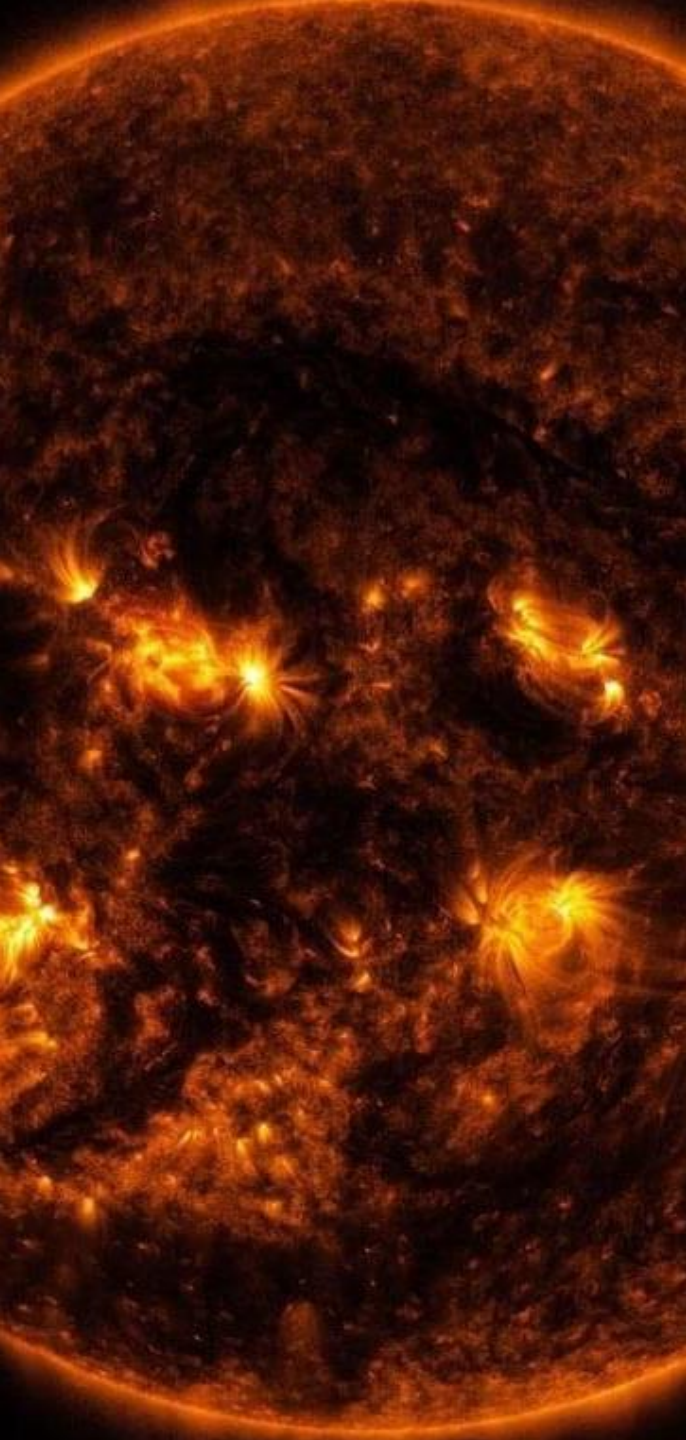


Pepe et al. 2021

STELLAR ACTIVITY

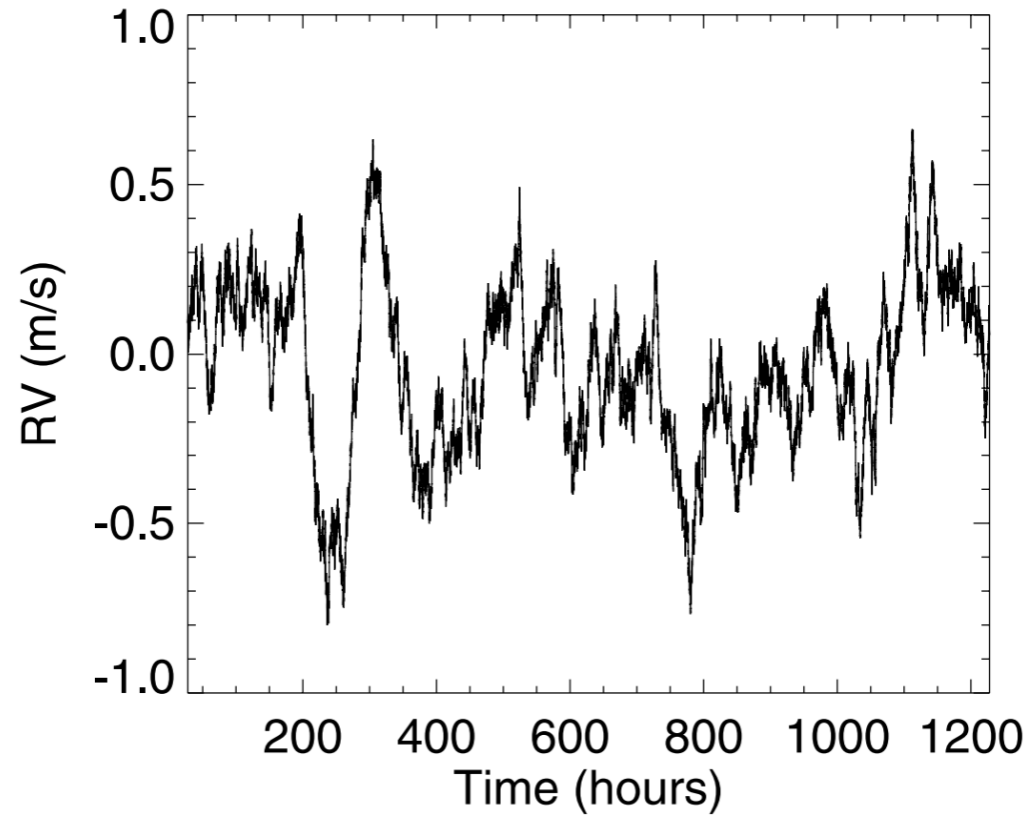
- Oscillations (P-modes): Minutes $\rightarrow$ Bin by exposure time



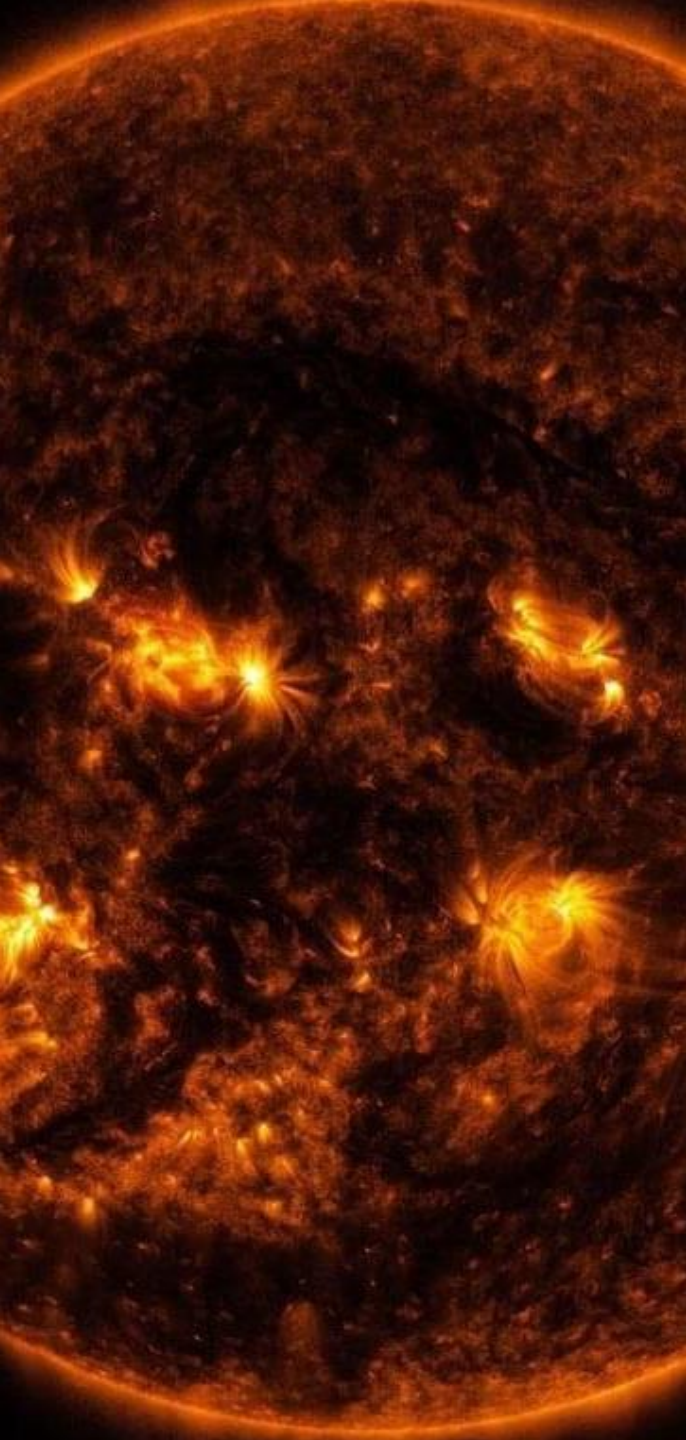


STELLAR ACTIVITY

- Granulation/Supergranulation: 1-2 hours to 1-2 days

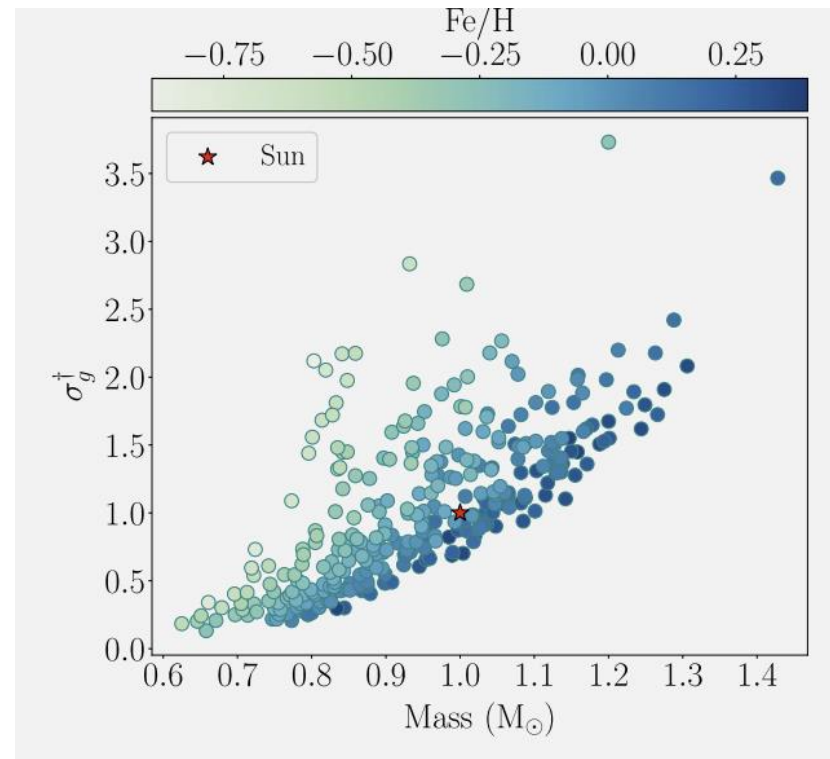


Meunier et al. 2015



STELLAR ACTIVITY

- Granulation/Supergranulation: 1-2 hours to 1-2 days



$$\log \sigma_g^\dagger = b_i (\log M)^2 + c_i (\log M) + d_i,$$

Metallicity bin	b_i	c_i	d_i
$[-0.89, -0.57]$	$-31.927^{+35.186}_{-43.919}$	$0.047^{+7.536}_{-9.379}$	$0.493^{+0.388}_{-0.482}$
$[-0.57, -0.25]$	$-18.706^{+12.526}_{-12.729}$	$2.262^{+1.79}_{-1.836}$	$0.314^{+0.07}_{-0.072}$
$[-0.25, 0.07]$	$-22.569^{+14.251}_{-12.978}$	$2.721^{+0.962}_{-0.871}$	$0.125^{+0.032}_{-0.034}$
$[0.07, 0.39]$	$-6.339^{+15.469}_{-14.778}$	$4.133^{+1.227}_{-1.319}$	$-0.06^{+0.057}_{-0.055}$

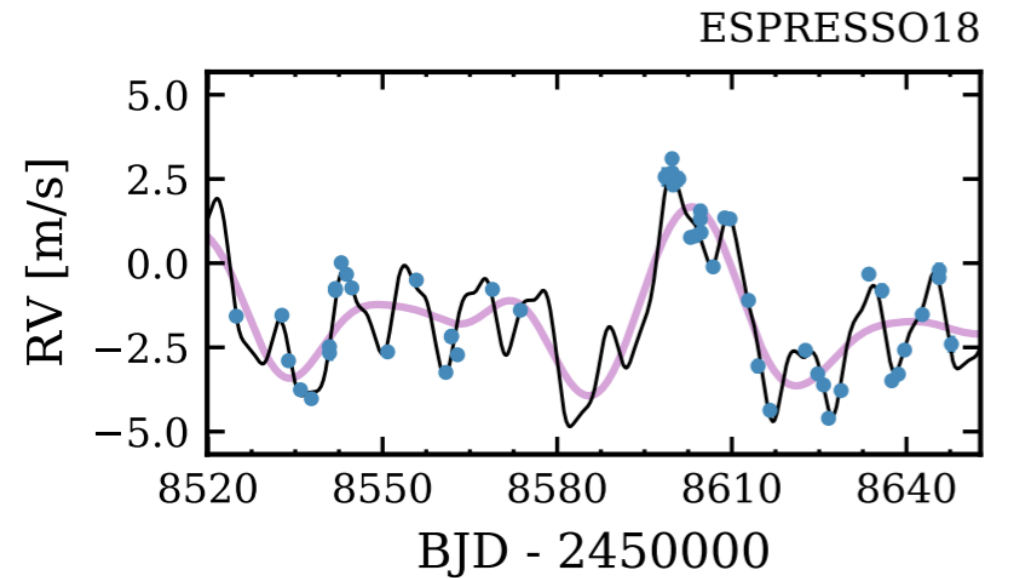
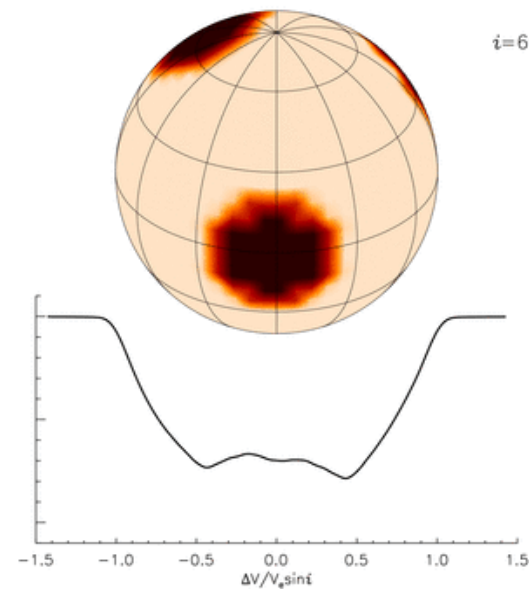
$$\sigma_g \text{ Sun} = 0.33 \text{ m/s}$$

Good enough approximation

F	0.5 – 1 m/s
G	0.3 m/s
K	0.1 m/s
M	0 m/s

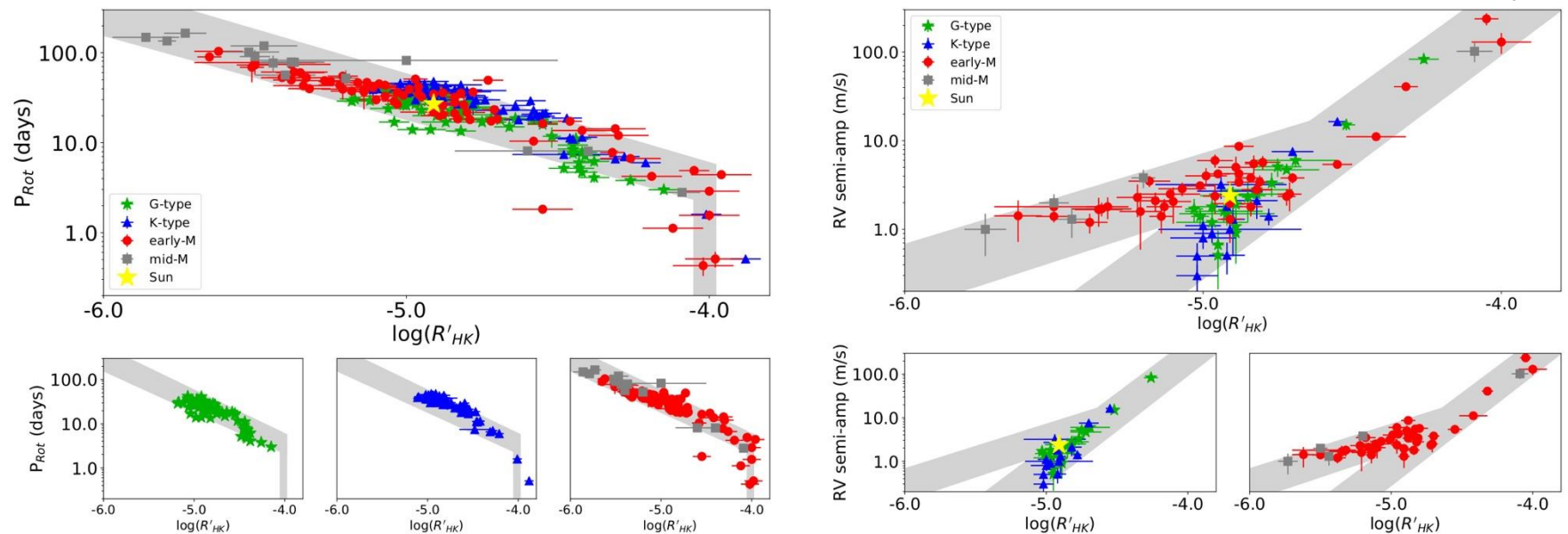
STELLAR ACTIVITY

- Stellar rotation: < 1 day to > 100 days



STELLAR ACTIVITY

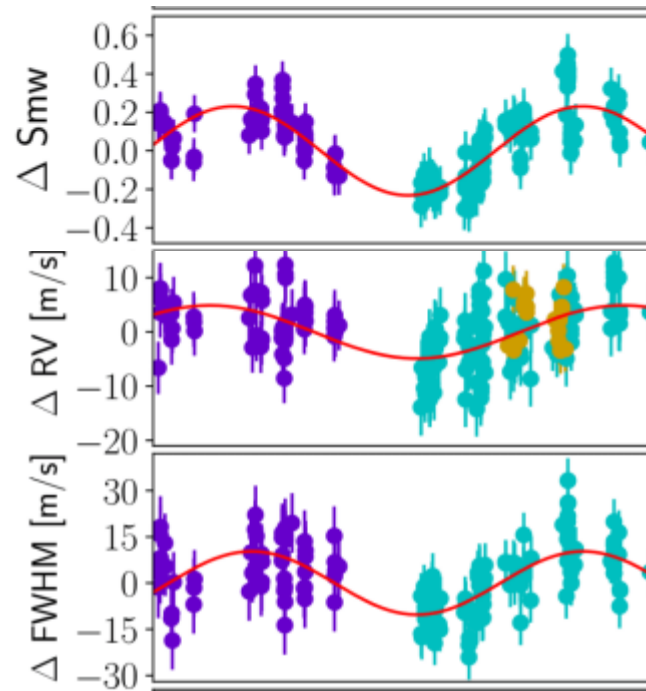
- Stellar rotation: < 1 day to > 100 days



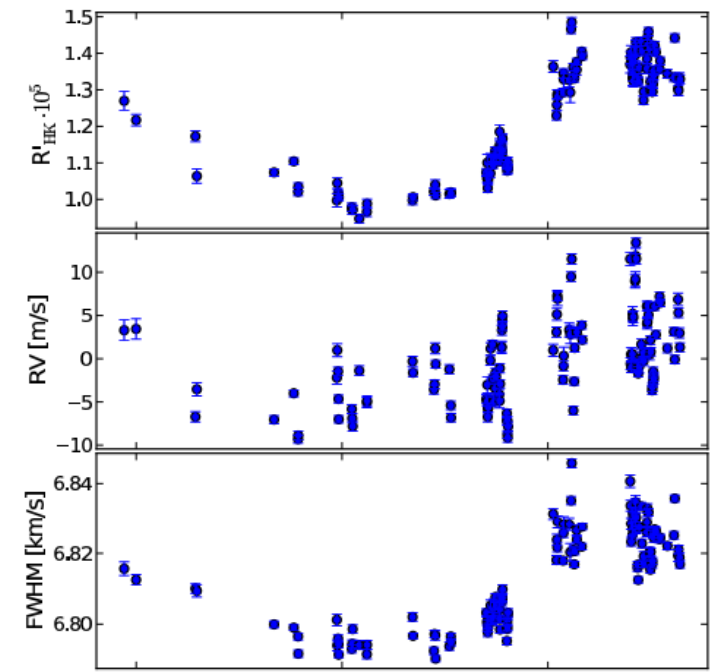
Based on Suárez Mascareño et al. 2015, 2016, 2017, 2018

STELLAR ACTIVITY

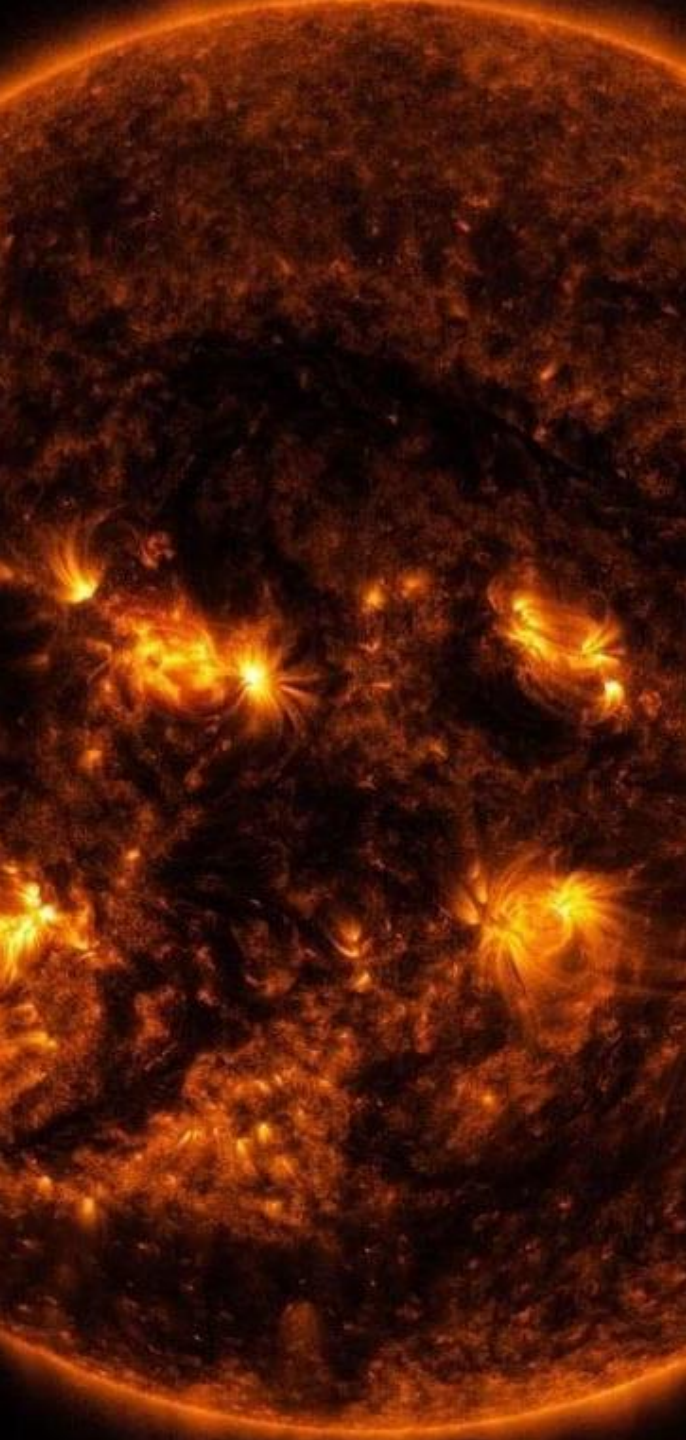
- Stellar cycles: Years



Toledo-Adr3n et al. 2021



Lovis et al. 2011



STELLAR ACTIVITY

$$\sigma^2_{\text{stellar}} = \sigma^2_{\text{oscillations}} + \sigma^2_{\text{granulation}} + \sigma^2_{\text{rotation}} + \sigma^2_{\text{cycles}}$$

- $\sigma_{\text{oscillations}} = 0 - 0.1$
- $\sigma_{\text{granulation}} = 0 - 1$ (depending on spectral type)
- $\sigma_{\text{rotation}} = K_{\text{rot}}/1.41$
- $\sigma_{\text{cycle}} = 0$ (for short proposals)

Sun-like star, 1 year of observations?

$$\sigma^2_{\text{stellar}} = 0.1^2_{\text{oscillations}} + 0.3^2_{\text{granulation}} + 1.8^2_{\text{rotation}} + 0^2_{\text{cycles}}$$

$$\sigma_{\text{stellar}} \sim 1.8 \text{ m/s}$$

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FIRST ORDER ESTIMATE

$$\sigma_{RV} = 1.8 \text{ m/s}$$

$$K = 1 \text{ m/s}$$

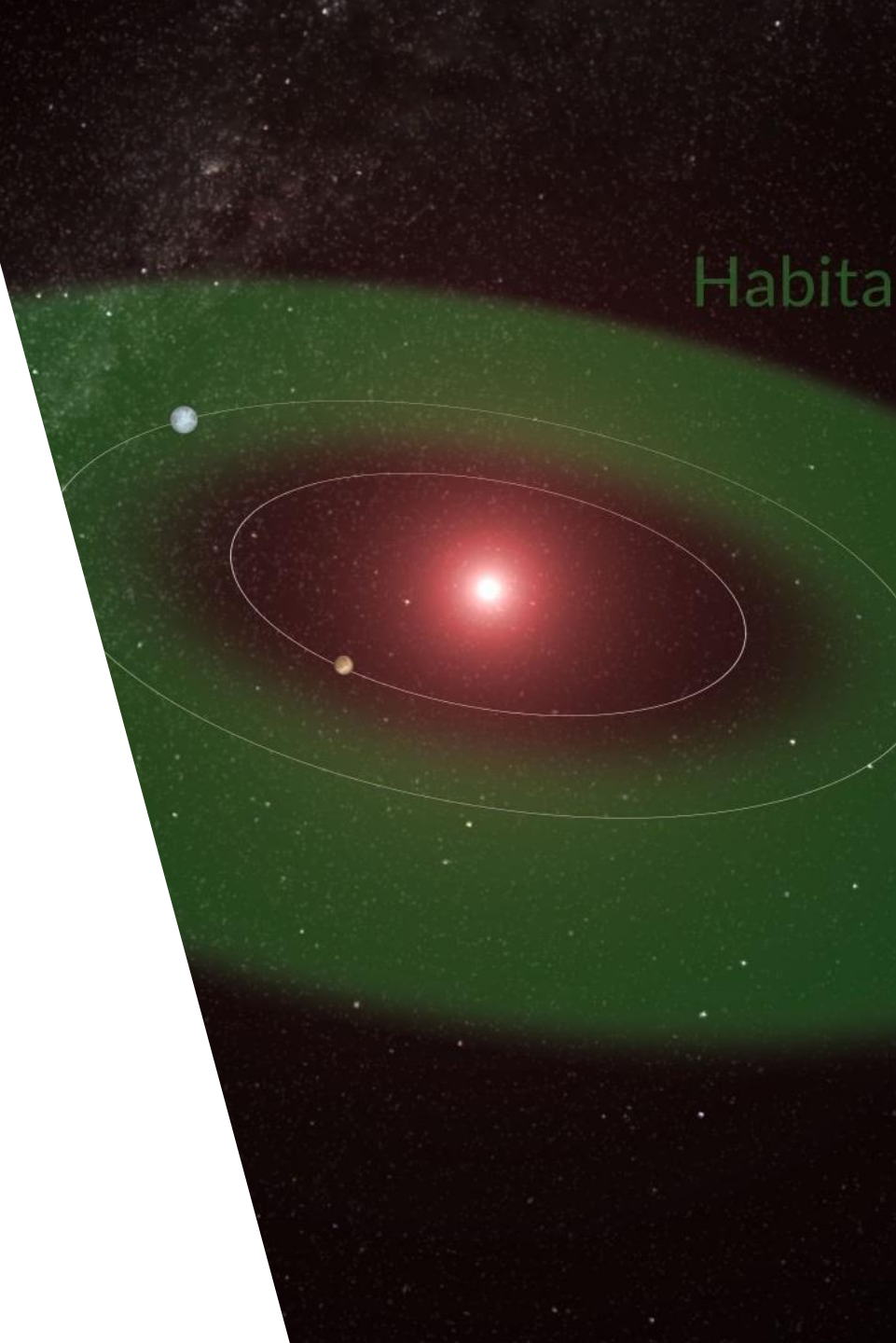
$$\sigma_K = 0.2 \text{ m/s}$$

$$f = \sigma_K / K = 0.2$$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{RV}}{\sqrt{N}},$$

$$N \simeq \frac{2}{f^2} \left(\frac{\sigma_{RV}}{K} \right)^2$$

$$N \sim 160 \text{ measurements}$$



Habita

FIRST ORDER ESTIMATE

$$\sigma_{RV} = 13 \text{ m/s}$$

$$K = 59 \text{ m/s}$$

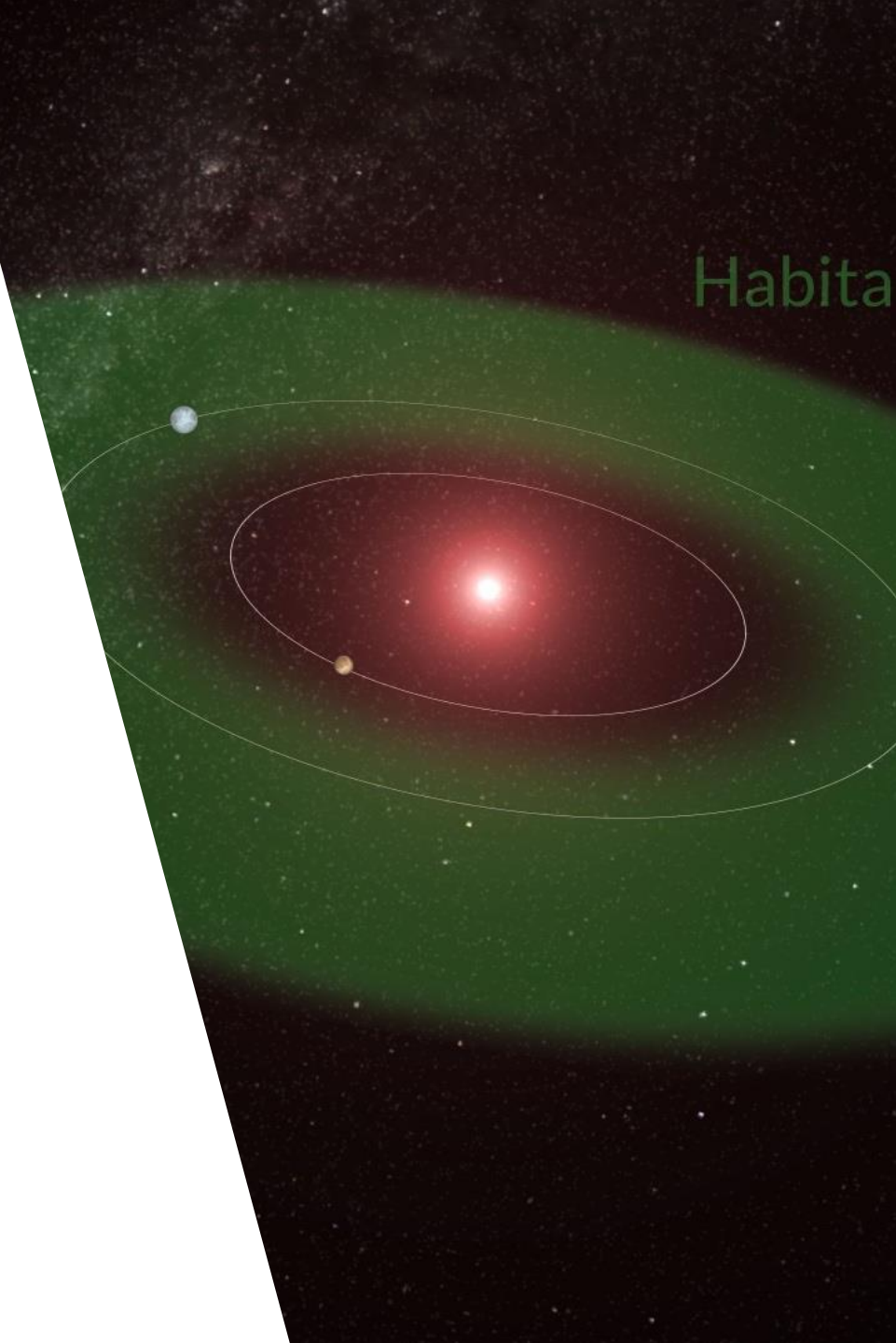
$$\sigma_K = 3 \text{ m/s}$$

$$f = \sigma_K / K = 0.05$$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{RV}}{\sqrt{N}},$$

$$N \simeq \frac{2}{f^2} \left(\frac{\sigma_{RV}}{K} \right)^2$$

$$N \sim 38 \text{ measurements}$$



FIRST ORDER ESTIMATE

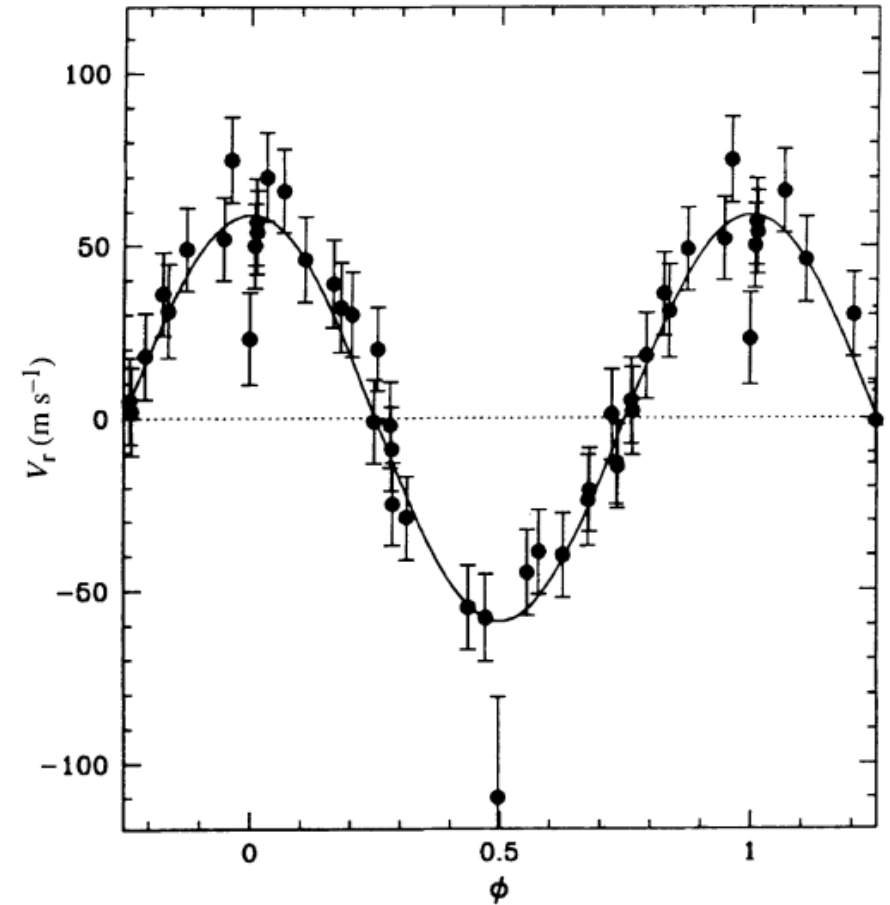
$$\begin{aligned}\sigma_{\text{RV}} &= 13 \text{ m/s} \\ K &= 59 \text{ m/s} \\ \sigma_K &= 3 \text{ m/s}\end{aligned}$$

$$f = \sigma_K / K = 0.05$$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{\text{RV}}}{\sqrt{N}},$$

$$N \simeq \frac{2}{f^2} \left(\frac{\sigma_{\text{RV}}}{K} \right)^2$$

$N \sim 38$ measurements

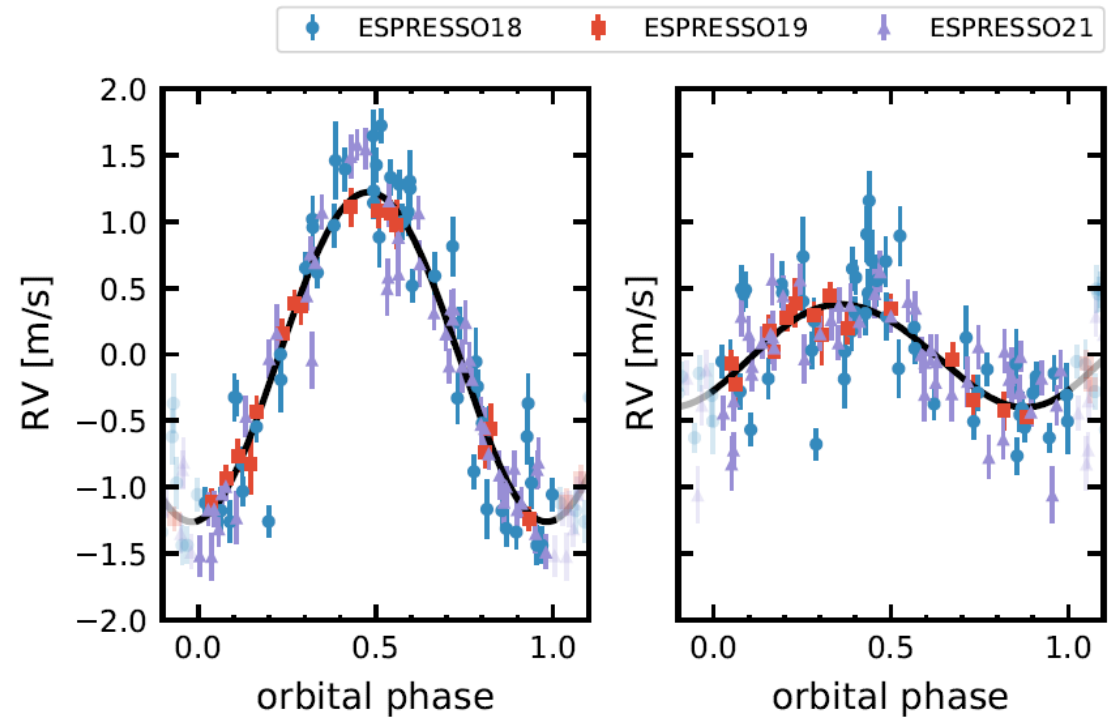


Mayor & Queloz needed 35

LIMITATIONS OF FIRST ORDER ESTIMATES

$\sigma_{\text{RV}} = 2.3 \text{ m/s}$
 $K_1 = 1.2 \text{ m/s}$
 $K_2 = 0.4 \text{ m/s}$
 $N = 117$
 $\sigma_{K \text{ meas}} = 0.07 \text{ m/s}$
 $\sigma_{K \text{ pred}} = 0.3 \text{ m/s}$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{\text{RV}}}{\sqrt{N}},$$



$N_{\text{pred}} \sim 2160$ measurements

LIMITATIONS OF FIRST ORDER ESTIMATES

$$\sigma_{\text{RV}} = 0.27 \text{ m/s}$$

$$K_1 = 1.2 \text{ m/s}$$

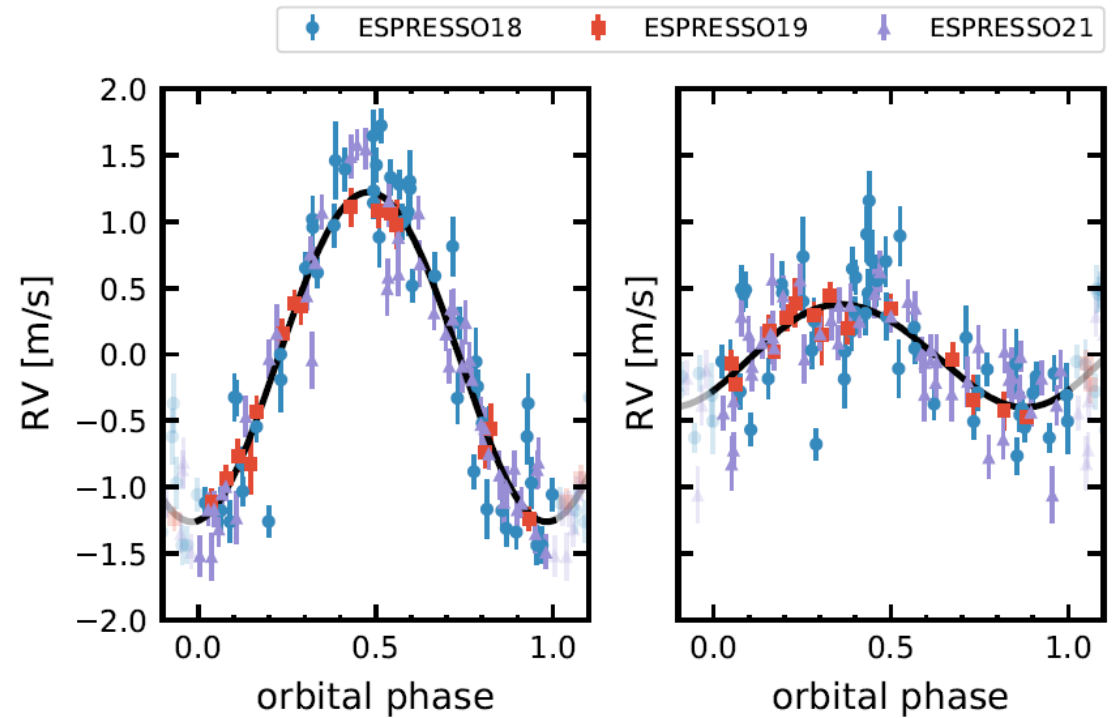
$$K_2 = 0.4 \text{ m/s}$$

$$N = 117$$

$$\sigma_{K \text{ meas}} = 0.07 \text{ m/s}$$

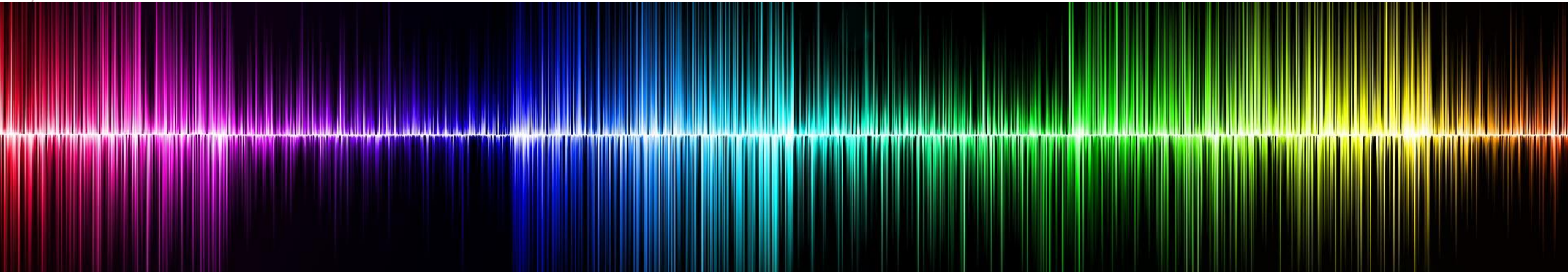
$$\sigma_{K \text{ pred}} = 0.035 \text{ m/s}$$

$$\sigma_K \simeq \frac{\sqrt{2} \sigma_{\text{RV}}}{\sqrt{N}},$$



$N_{\text{pred}} \sim 27$ measurements

*CAN WE MAKE A BETTER
ESTIMATION?*

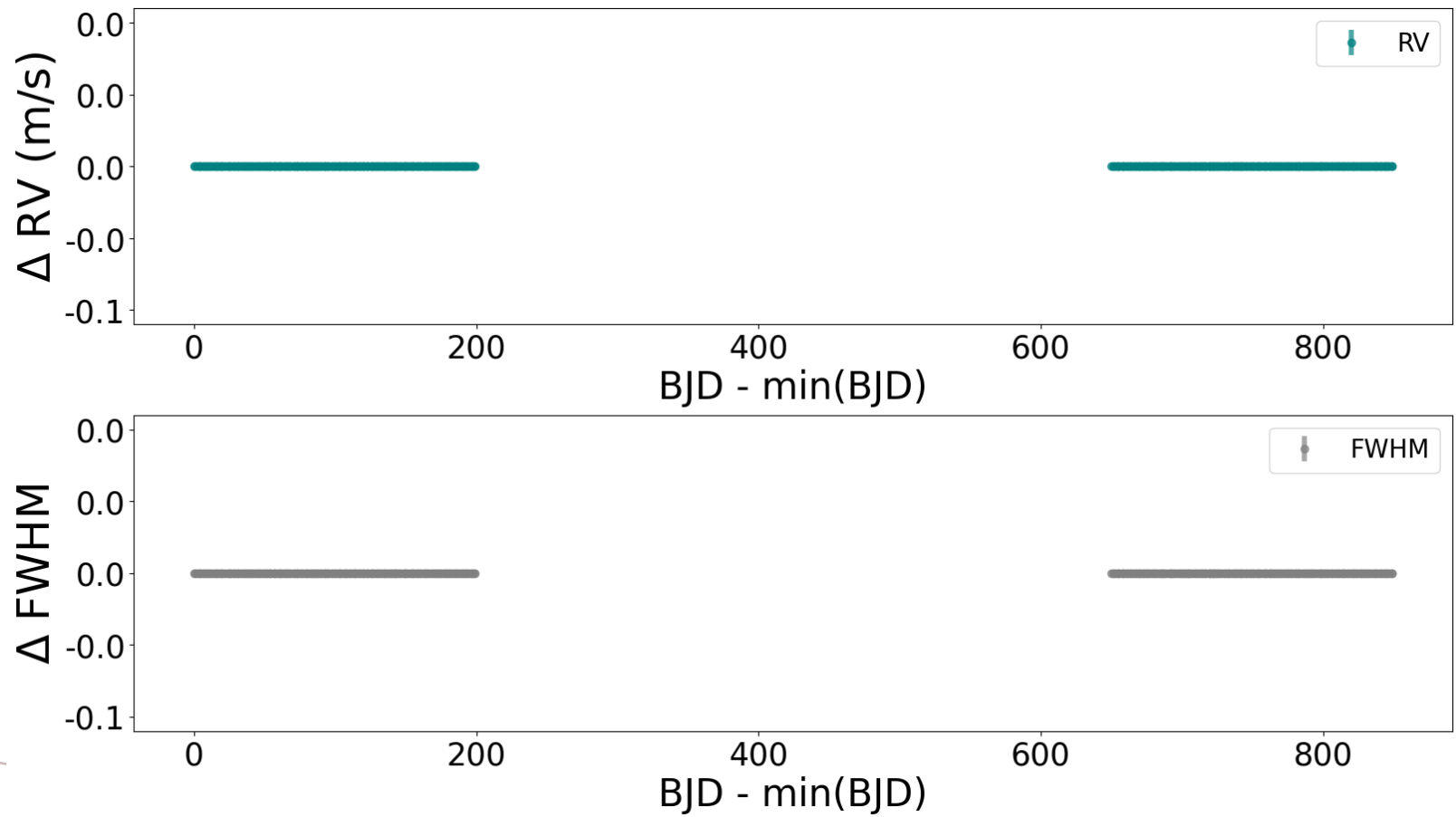


CAN WE MAKE A BETTER ESTIMATION?

1. Yes
2. It's not that hard
3. I promise

We don't need to actually mimic the star. Only produce data that (roughly) behaves the same.

I - TIMESTAMPS



1 point per night, at a random hour,
for 2 years of visibility

I - TIMESTAMPS

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OPEN ACCESS

<https://doi.org/10.3847/1538-3881/aaa47e>



astroplan: An Open Source Observation Planning Package in Python

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astroplan

docs passing arXiv 1709.03913 CI Tests passing powered by AstroPy pypi v0.10.1

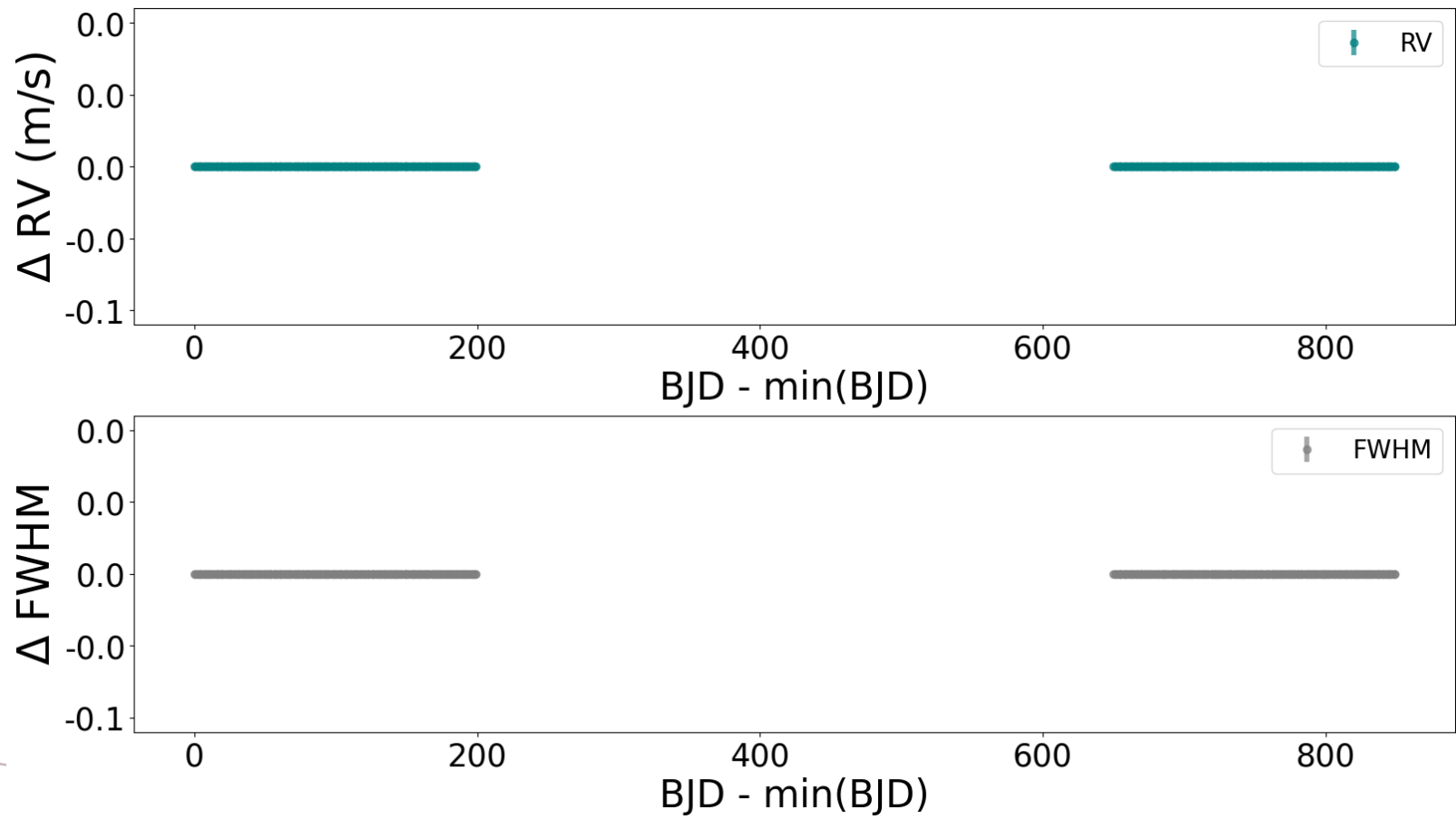
- Code: <https://github.com/astroplan/astroplan>
- Docs: <https://astroplan.readthedocs.io/>
- License: BSD-3 Clause

astroplan is an open-source observation planning package for astronomers that can help you plan for everything but the clouds.

It is an [Astropy affiliated package](#) that seeks to make your life as an observational astronomer a little less infuriating.

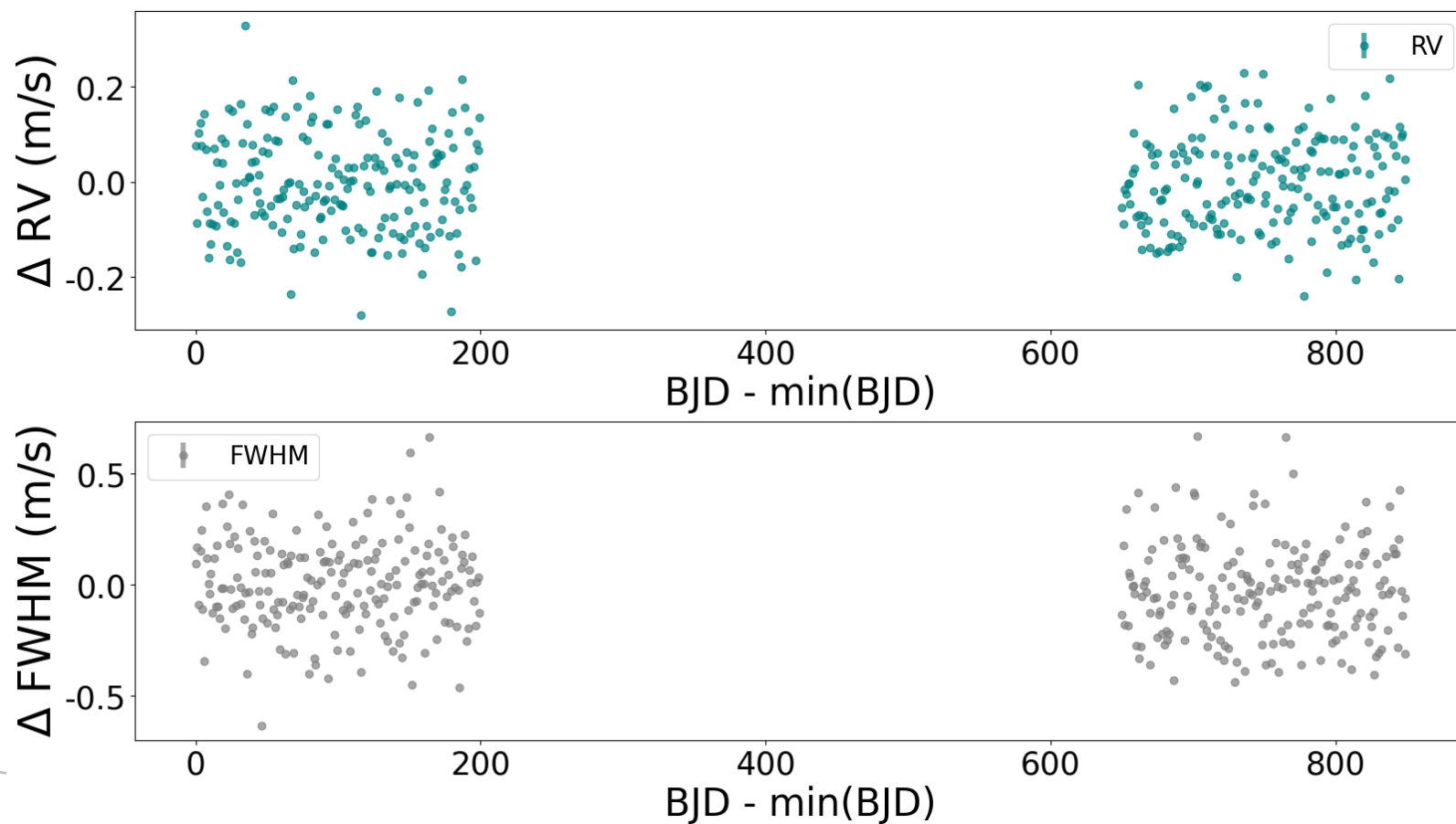
Contributions welcome!

I - TIMESTAMPS



1 point per night, at a random hour,
for 2 years of visibility

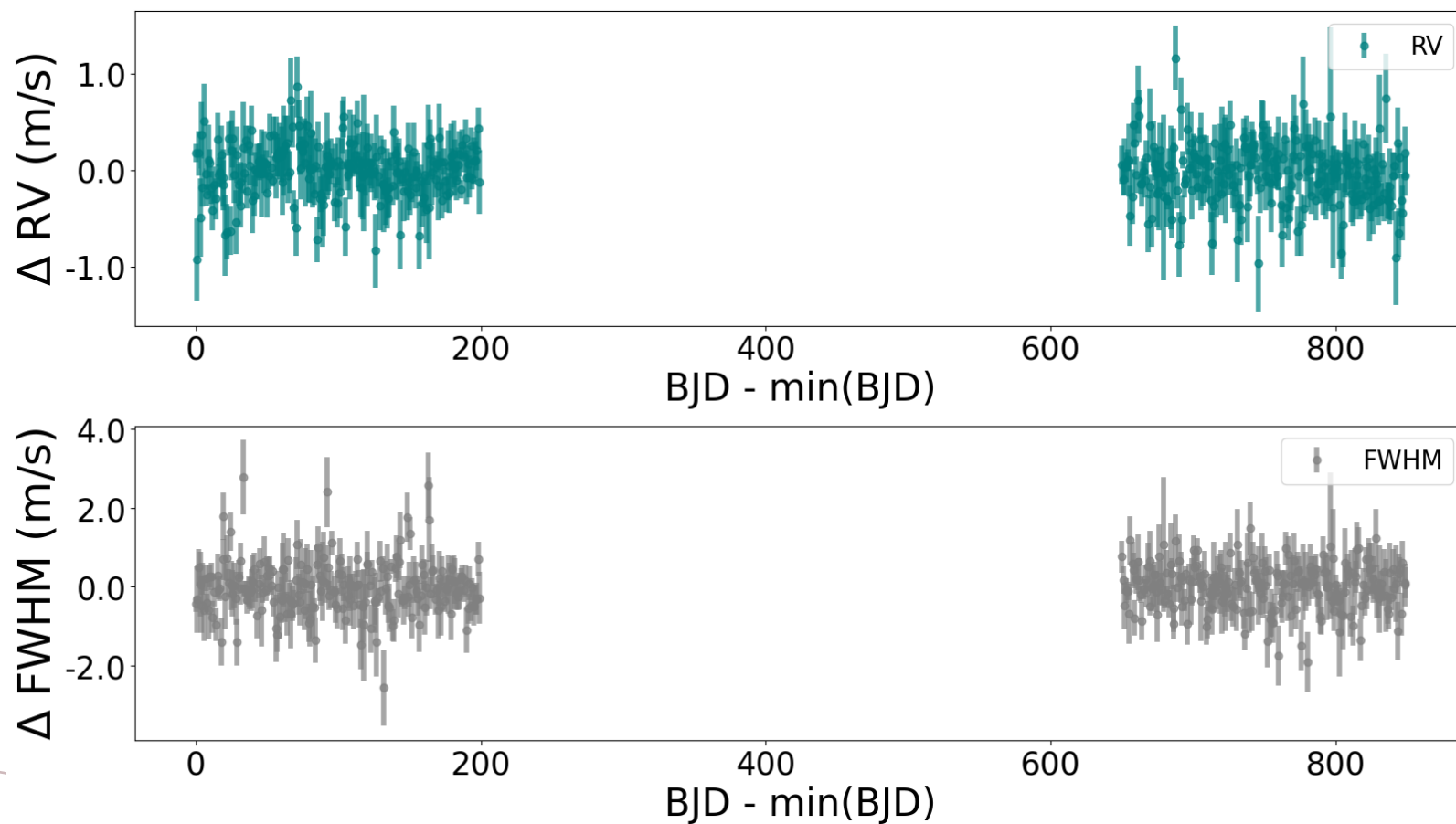
II - INSTRUMENT



ESPRESSO:

- 10 cm/s RMS RV noise
- 20 cm/s RMS FWHM noise

II – PHOTON NOISE



ESPRESSO:

- 25 cm/s RMS RV noise
- 50 cm/s RMS FWHM noise

II – ACTIVITY

```
prot = 80+np.random.uniform(0,10)
krot = np.random.normal(2.3*1.4,0.3)
t01 = np.random.uniform(0,prot)
t02 = np.random.uniform(0,prot*0.5)
t03 = np.random.uniform(0,prot*0.33)
k2 = np.random.uniform(0,0.6)
k3 = np.random.uniform(0,0.3)

trend = 1-0.3*np.abs(np.sin(2*np.pi*(x_orig-45)/250))

y_act = -krot*np.sin(2*np.pi*(x_orig-t01)/prot) \
        - k2*krot*np.sin(4*np.pi*(x_orig-t02)/prot) \
        - k3*krot*np.sin(6*np.pi*(x_orig-t03)/prot)
y_act = y_act*trend
y_act = y_act/np.std(y_act)*10

grad_act = np.gradient(y_act)/np.std(np.gradient(y_act))
lin_act = y_act/np.std(y_act)

y_act_rv = 0.9*grad_act - 0.1*lin_act
```

“Base activity”: Sum of 3 sinusoids (P, P/2, P/3) with somewhat random parameters.

Activity in RV: FF' projection into RV.

Proxima:

- 85 days rotation
- 2.3 m/s RV RMS
- 10 m/s FWHM RMS

II – ACTIVITY

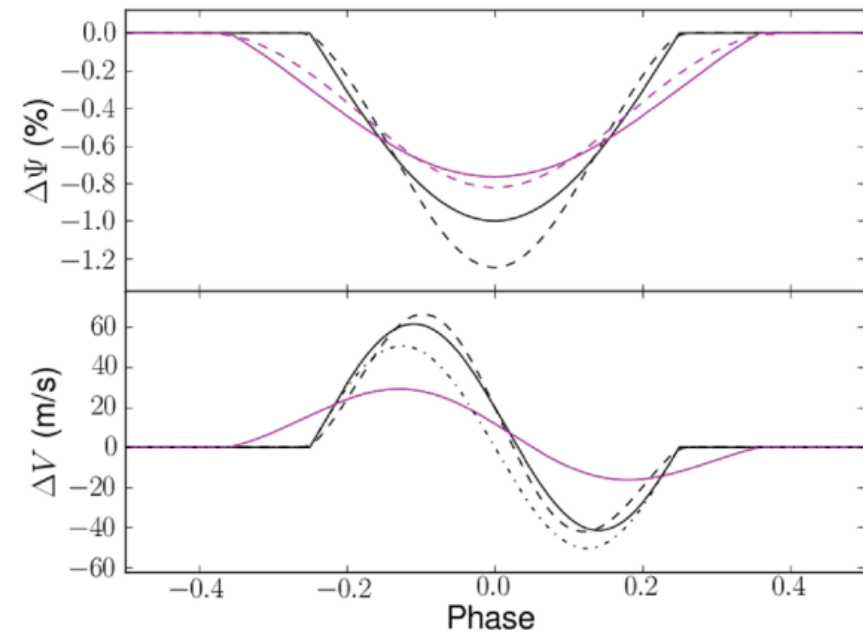
```
prot = 80+np.random.uniform(0,10)
krot = np.random.normal(2.3*1.4,0.3)
t01 = np.random.uniform(0,prot)
t02 = np.random.uniform(0,prot*0.5)
t03 = np.random.uniform(0,prot*0.33)
k2 = np.random.uniform(0,0.6)
k3 = np.random.uniform(0,0.3)

trend = 1-0.3*np.abs(np.sin(2*np.pi*(x_orig-45)/250))

y_act = -krot*np.sin(2*np.pi*(x_orig-t01)/prot) \
        - k2*krot*np.sin(4*np.pi*(x_orig-t02)/prot) \
        - k3*krot*np.sin(6*np.pi*(x_orig-t03)/prot)
y_act = y_act*trend
y_act = y_act/np.std(y_act)*10

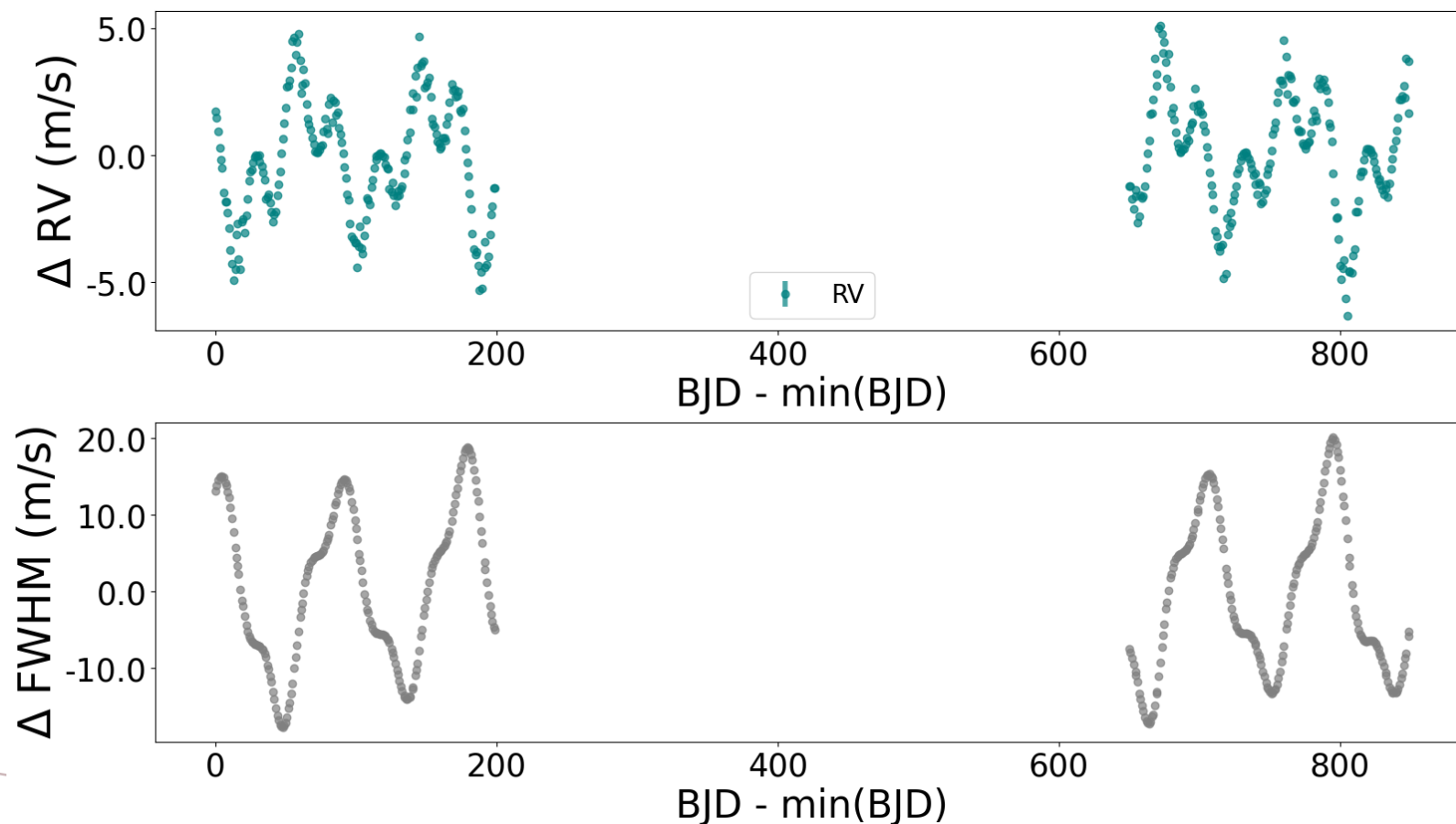
grad_act = np.gradient(y_act)/np.std(np.gradient(y_act))
lin_act = y_act/np.std(y_act)

y_act_rv = 0.9*grad_act - 0.1*lin_act
```



Aigrain et al. 2011

II – ACTIVITY

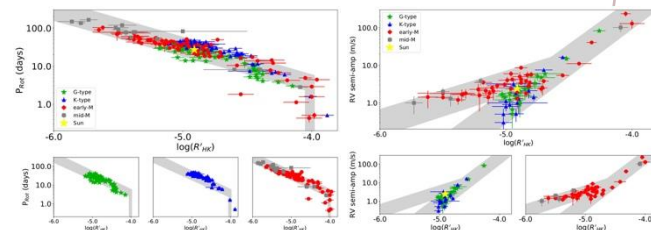


Proxima:

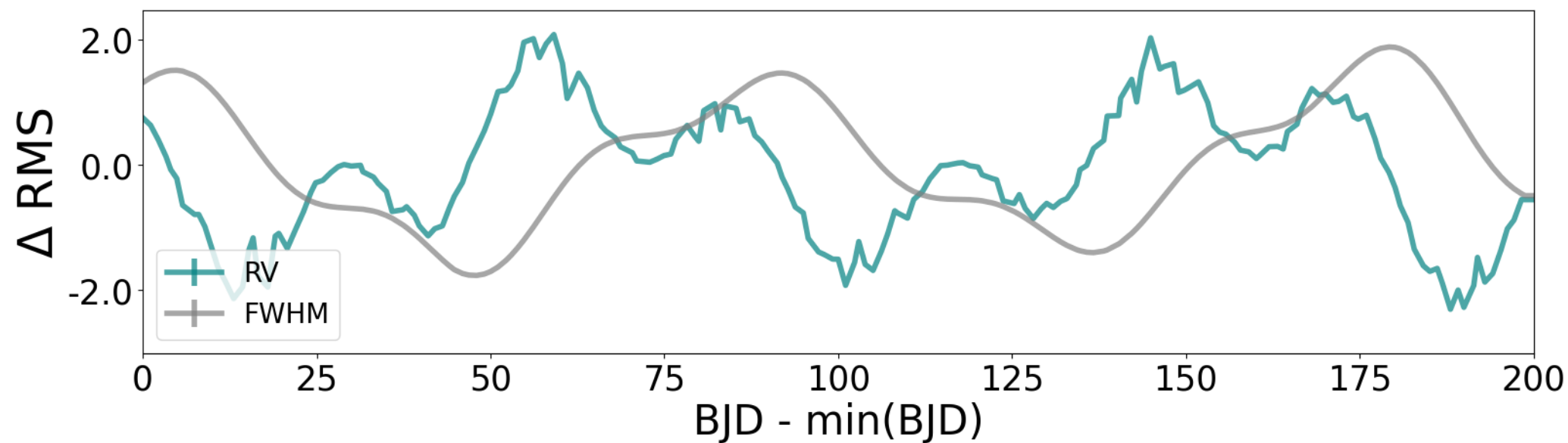
- 85 days rotation
- 2.3 m/s RV RMS
- 10 m/s FWHM RMS

What if I don't know the values from my star?

- Use rotation-activity relationships from Rhk or Prot

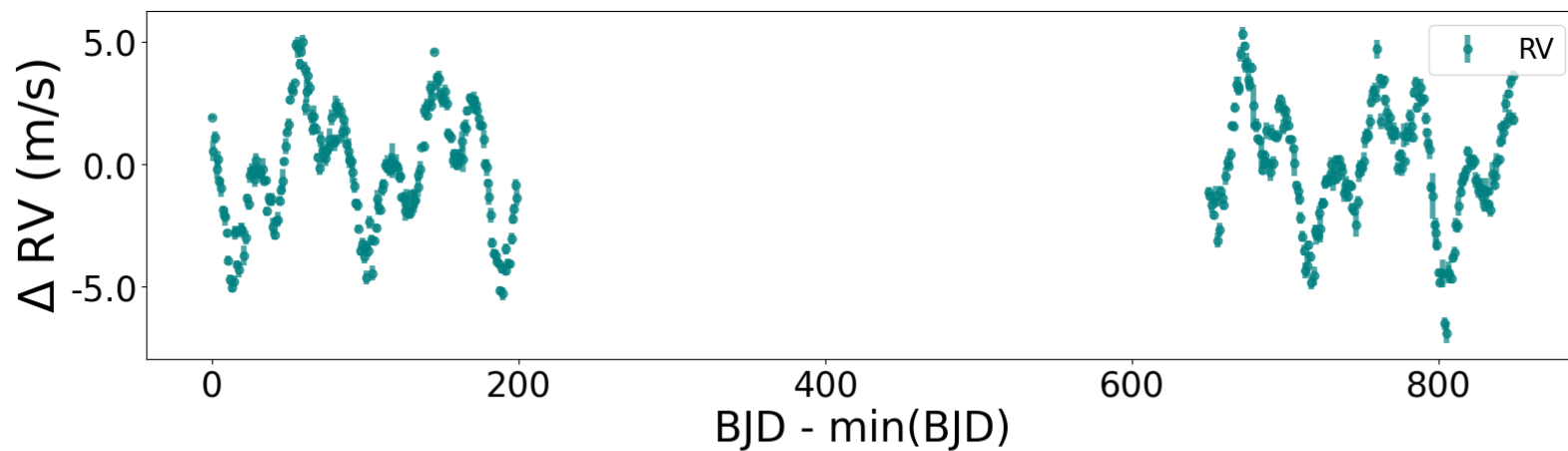


II – ACTIVITY

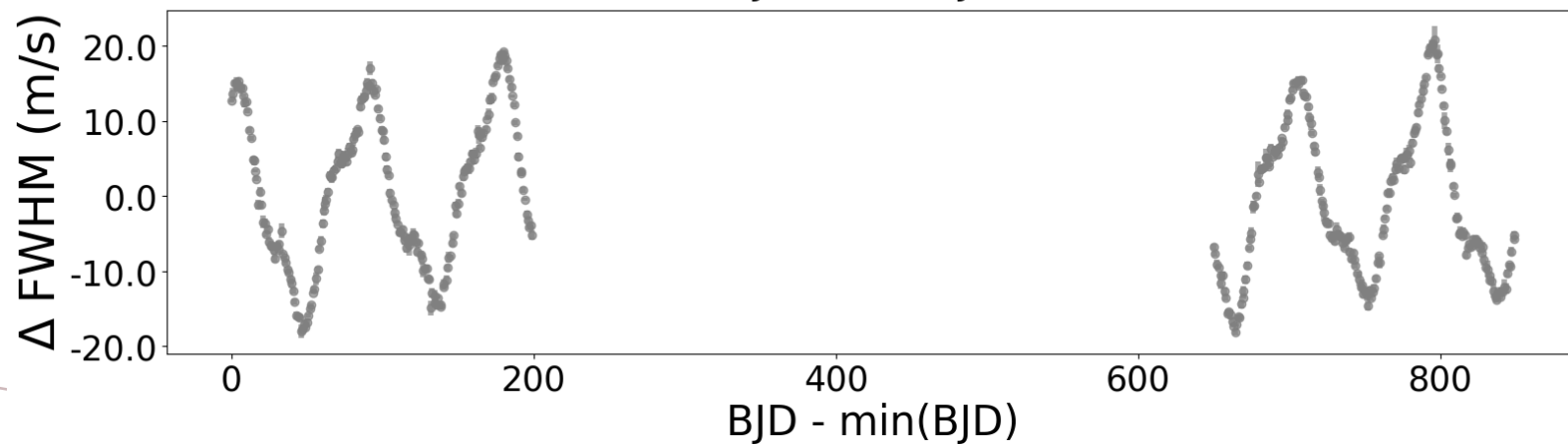


Using FF' concept implicitly creates phase lags and higher complexity in RV

III – ACTIVITY+NOISE

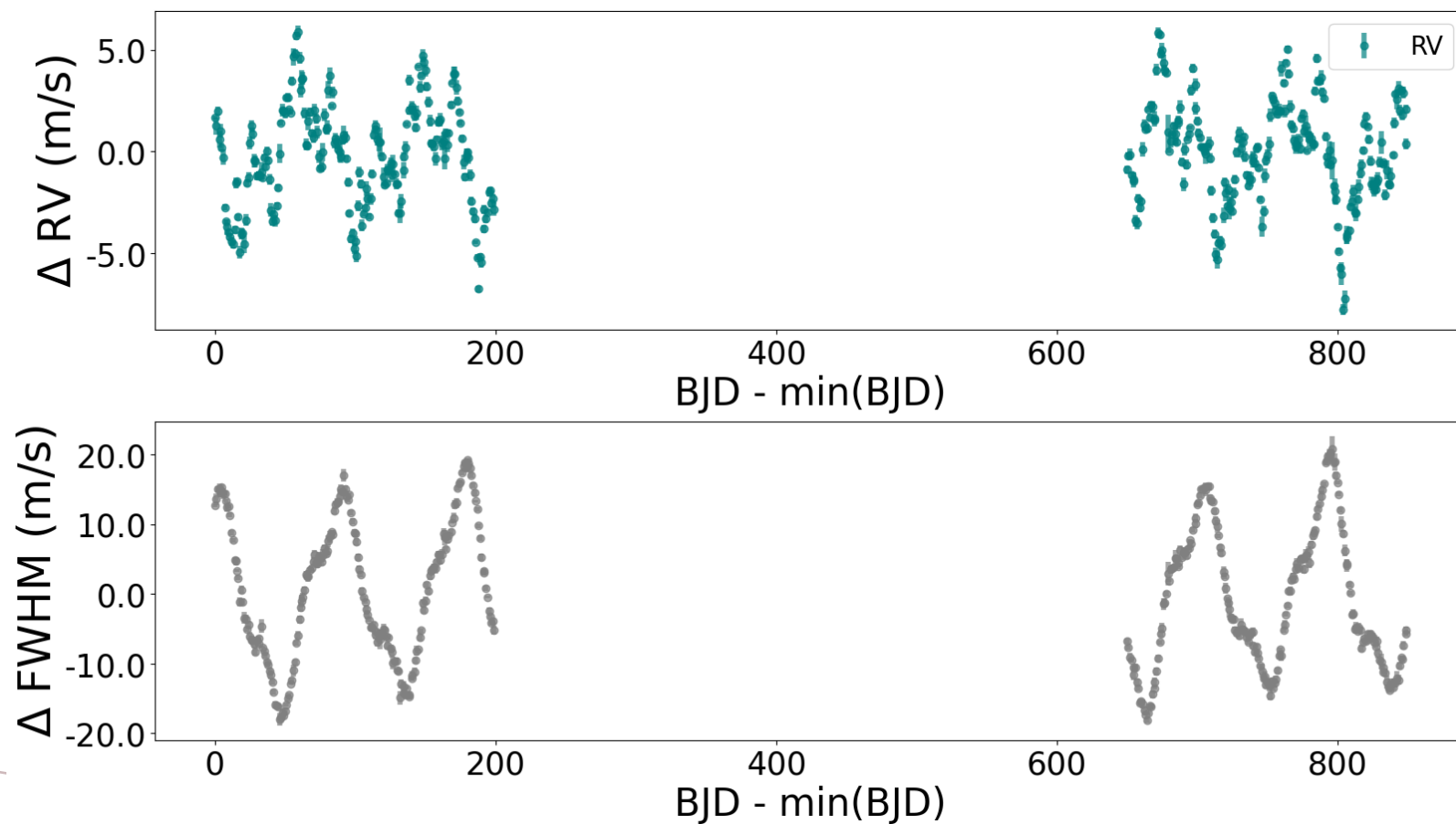


For bright targets, activity will always dominate over instrument and photon noise.



For faint targets it might not be the case.

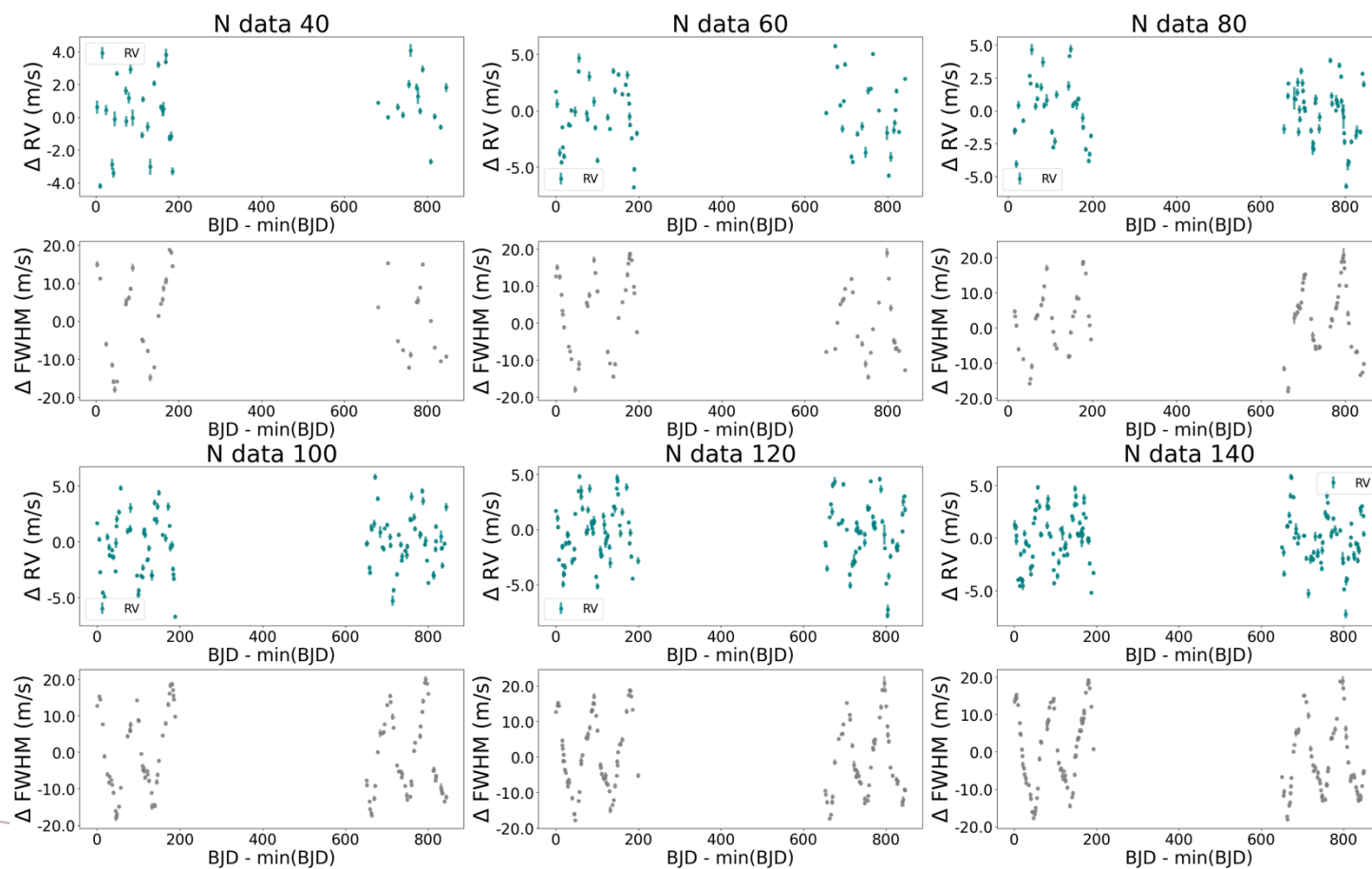
IV – ACTIVITY+NOISE+PLANETS



Two planets added.

- Similar to Proxima b and d.

V – FINAL SAMPLING



Randomly selected timestamps
between 40 and 140
measurements.

VI – MODELLING

```
T      = [jd,jd]
Y      = [fwhm,rv]
Yerr   = [sfwhm,erv]
t_full, y_full, yerr_full, series_index = cov.merge_series(T, Y, Yerr)

GP1 = term.MultiSeriesKernel(term.SHOKernel(1, 43,43,1), series_index, \
                               [1, 1], \
                               [0 ,1])
GP2 = term.MultiSeriesKernel(term.SHOKernel(1, 43,43,1), series_index, \
                               [1, 1], \
                               [0 ,1])

C = cov.Cov(t_full,
             err      = term.Error(yerr_full),
             fwhm_jit = term.InstrumentJitter(series_index[0], 1.0),
             rv_jit   = term.InstrumentJitter(series_index[1], 1.0),
             term1 = GP1,term2=GP2)

params_gp_names = C.param
```

```
priors = [[r"Log Jit FWHM","N",2.3,2.3],
          [r"Log Jit RV","N",0.8,0.8],
          [r"V0 FWHM","N",0.0,30],
          [r"V0 RV","N",0.0,10],
          [r"Period Rot (d)","N",85,20],
          [r"Log TS","U",3,7],
          [r"Log Scale","U",-5,10],
          [r"A1 FWHM","U",0,100],
          [r"A1 RV","U",-50,100],
          [r"B1 RV","U",-200,400],
          [r"K 1 (m/s)","U",0,5],
          [r"Period 1 (d)","N",11.2,0.1],
          [r"PH 1","U",0,1],
          [r"K 2 (m/s)","U",0,5],
          [r"Period 2 (d)","N",5.12,0.05],
          [r"PH 2","U",0,1]]

priors_ns,priors_print,labels = prepare_priors(priors)

pool = Pool()
cpu = multiprocessing.cpu_count() - 2

ndim = len(priors_ns)
nlive = ndim*20
dlogz_stop=0.01
queue = int(cpu*2)

print("N dim",ndim)
print("N live",nlive)
print("Queue size",queue, "Queue / cpu",queue/cpu)

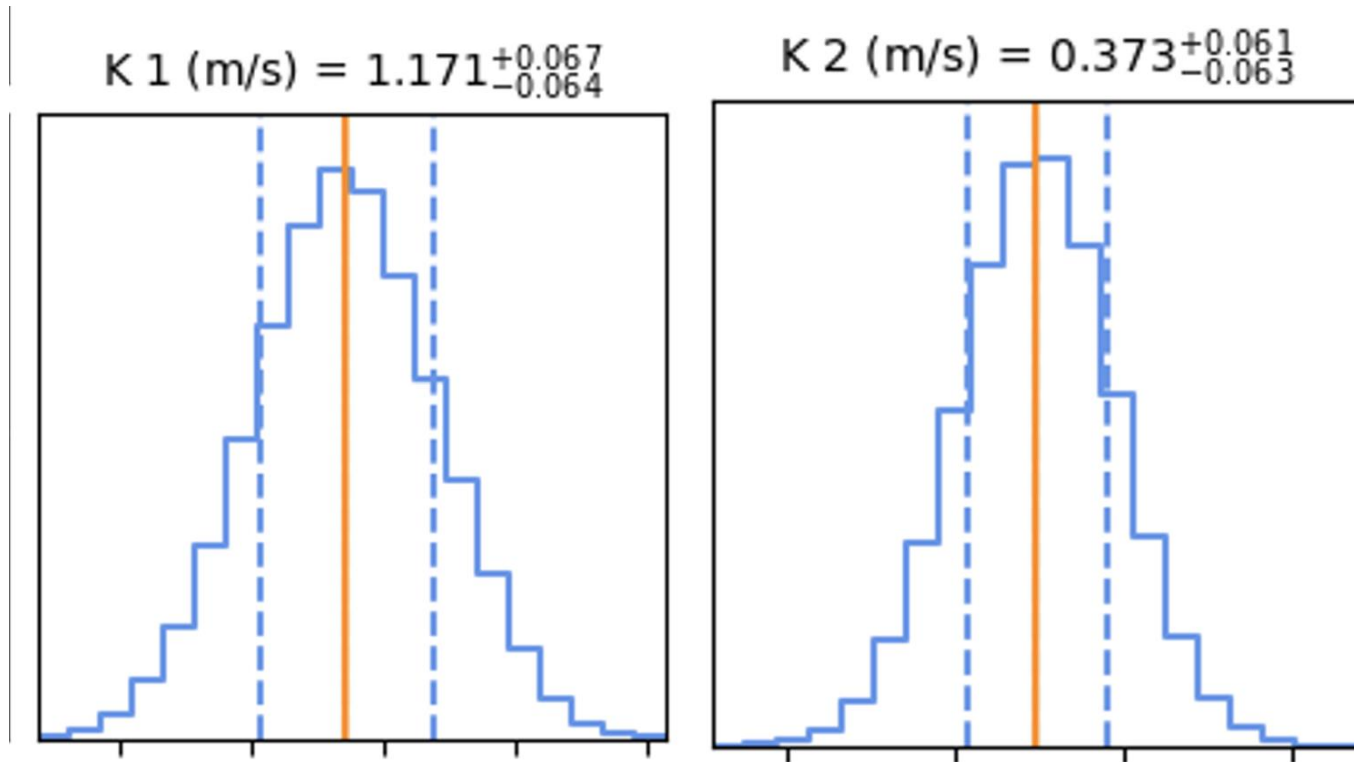
sampler = dynesty.NestedSampler(loglike, prior_transform, ndim,nlive=nlive, pool=pool,queue_size=queue,sample="rslice")

print(sampler.citations)

# Run the sampler
sampler.run_nested(dlogz=dlogz_stop)
pool.close()
```

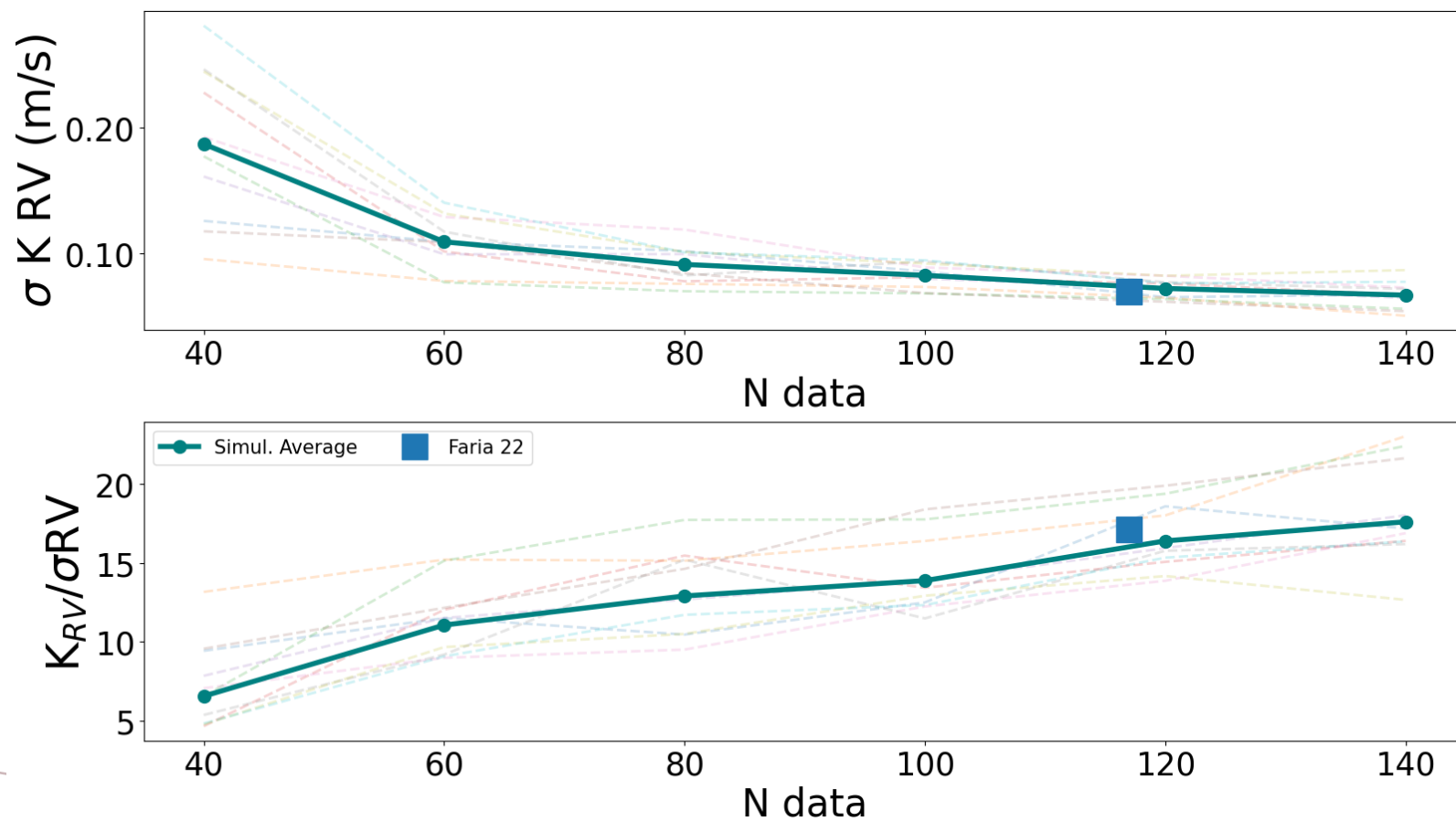
Model the data the simulated data the same way you would model the real data.

VII – RESULTS



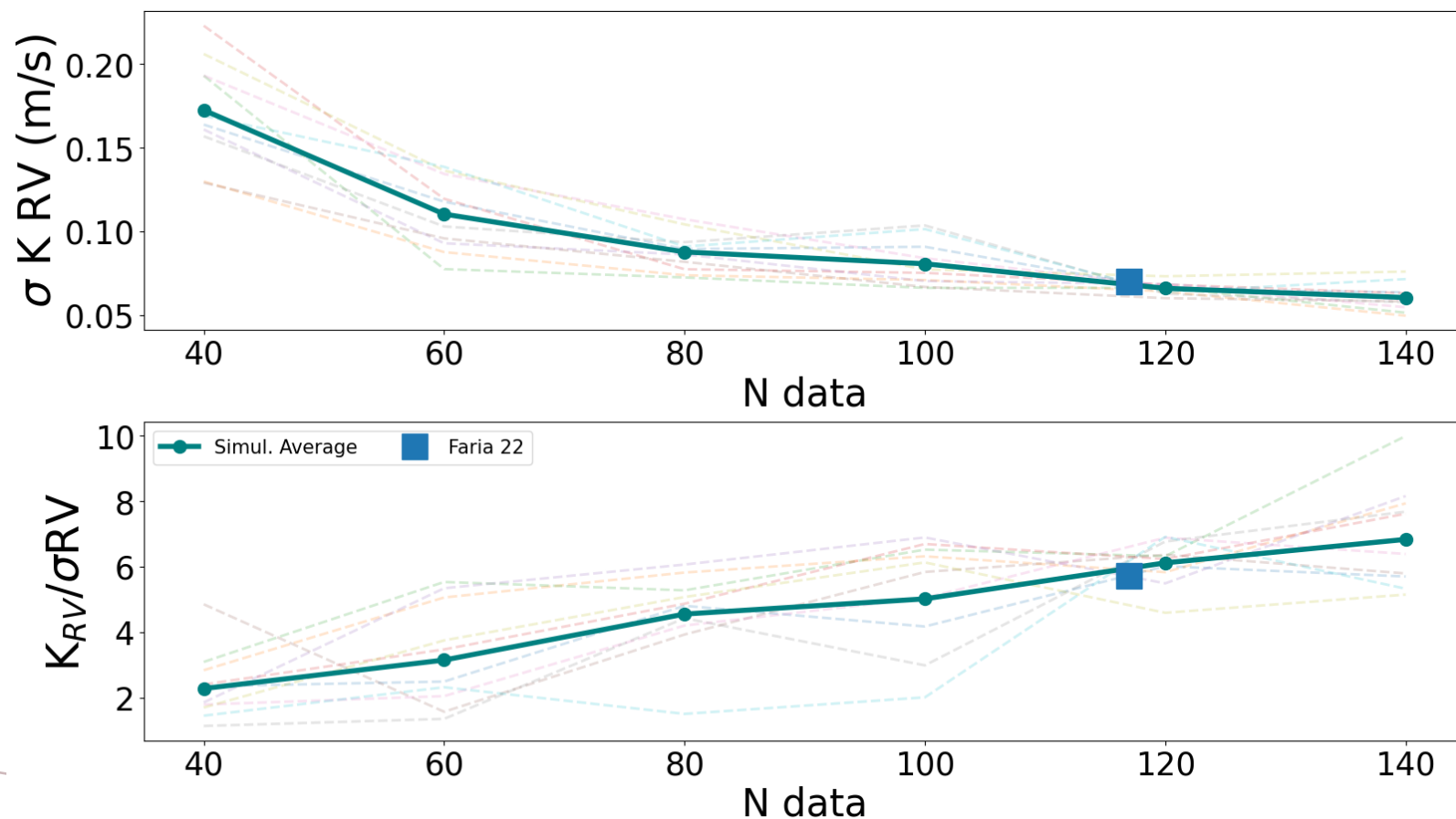
Consistent with Faria et al. 22

VII – RESULTS



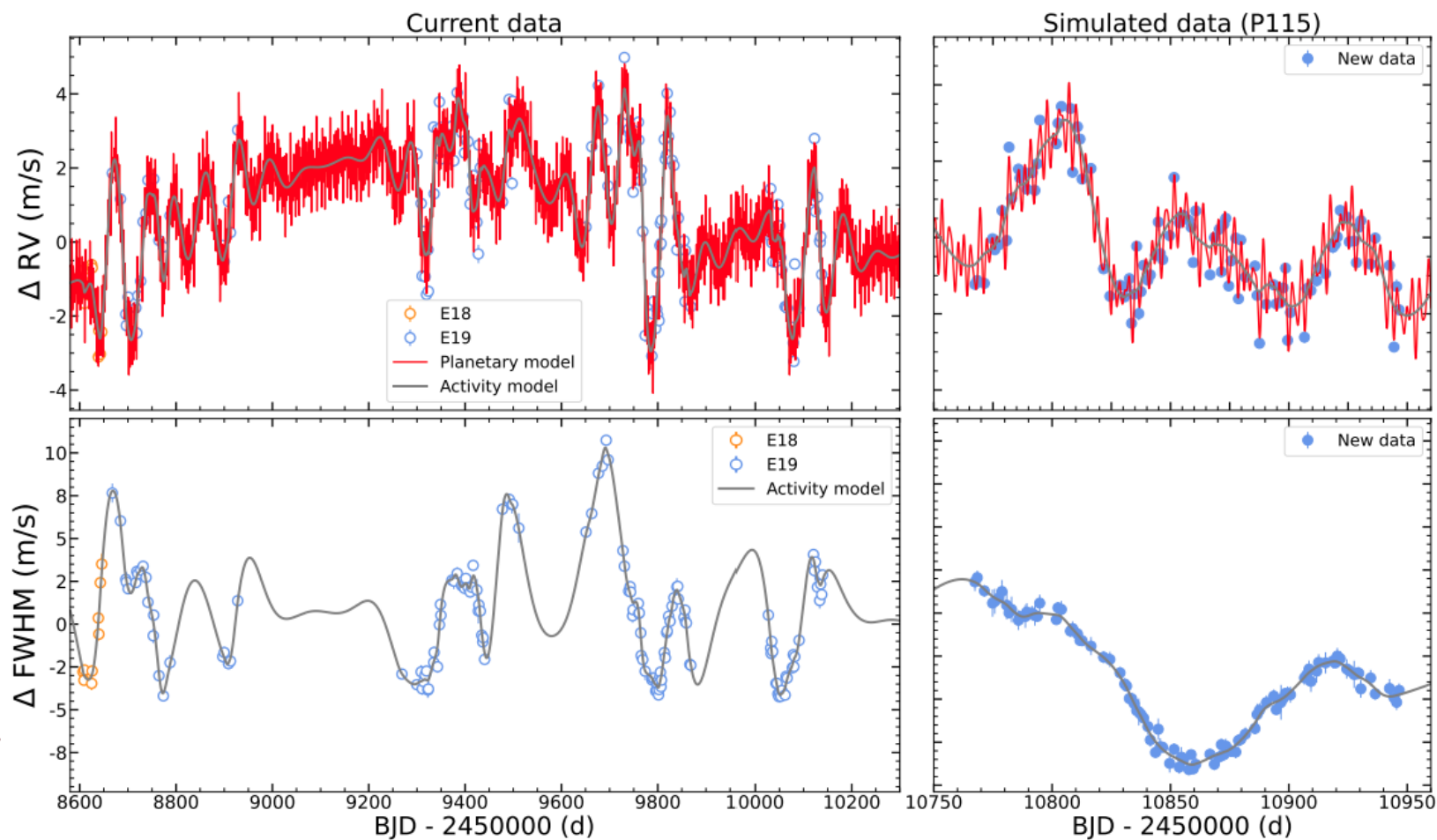
For a Proxima b-like signal, we predict the right precision at the right number of observations.

VII – RESULTS



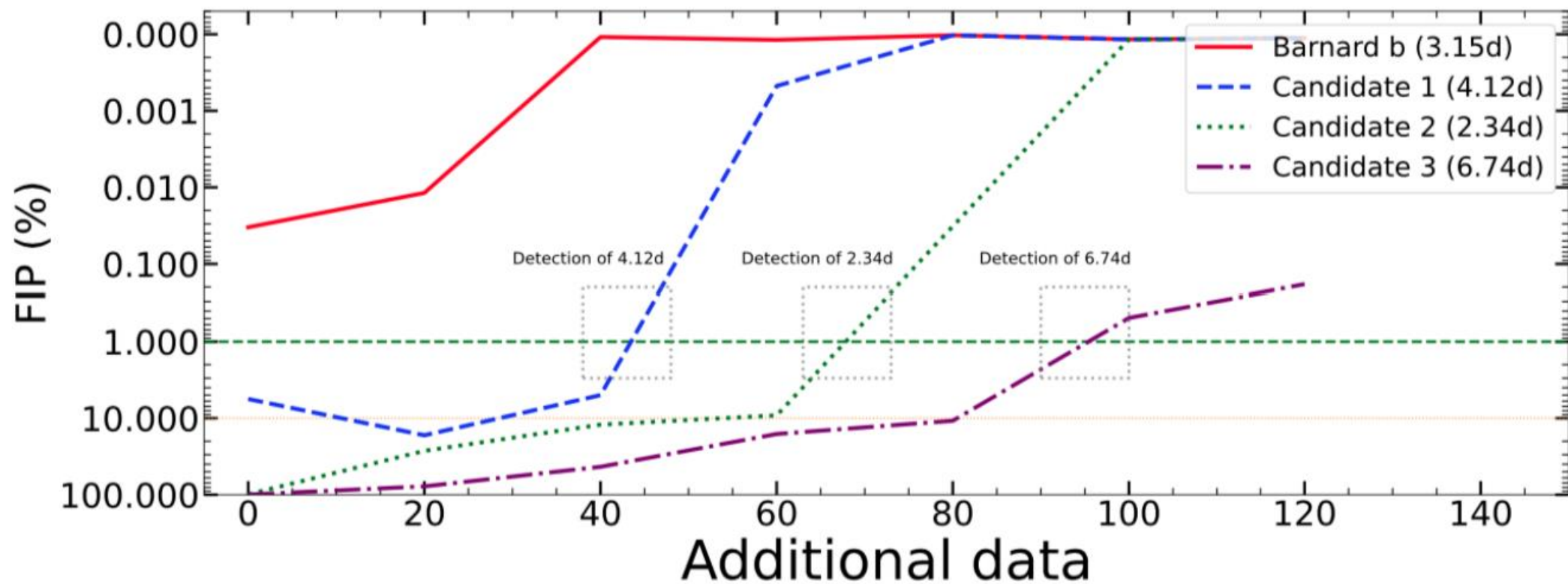
For a Proxima d-like signal, we predict the right precision at the right number of observations.

VIII – EXAMPLES



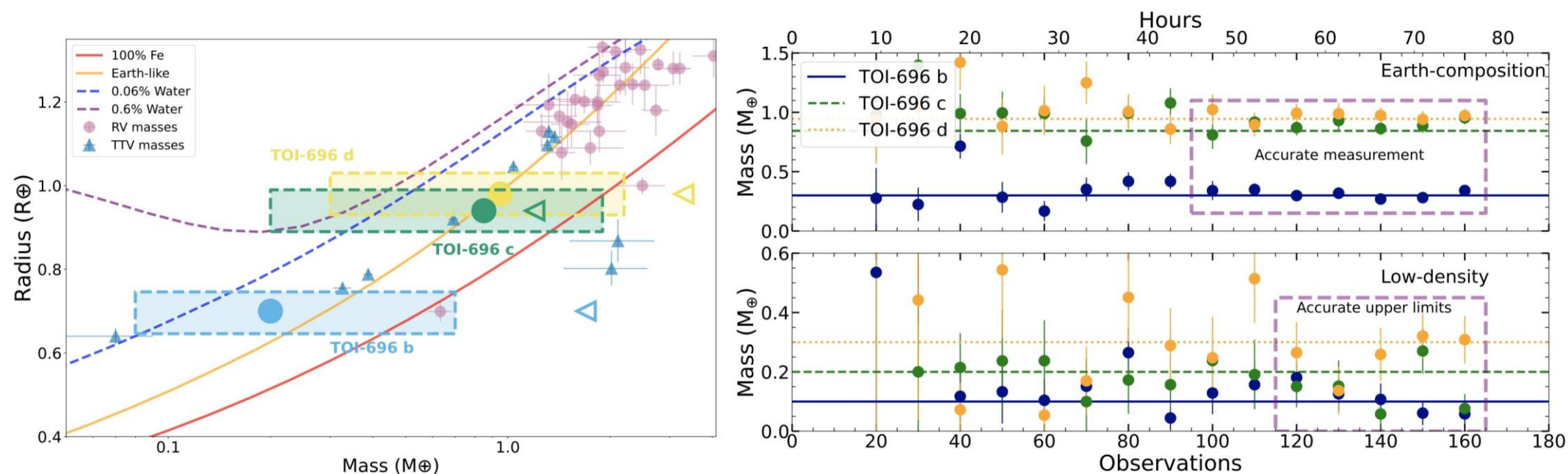
Simulated data of Barnard's star with ESPRESSO (approved proposal P115)

VIII – EXAMPLES



Simulated results of a campaign on Barnard's star with ESPRESSO (approved proposal P115)

VIII – EXAMPLES



Simulated results of a campaign on LHS 1678 with ESPRESSO (approved proposal P116)

IV – THE CURSE OF TIME-SERIES

Time-series data are difficult to obtain.

- *Many observatories are not efficient at taking time series*
- *Weather losses and conflicts with other programs will create unexpected gaps in the data*
- *Low priority ranking affects time-series data more than other types of observations*
- *You are rarely getting as much data as you are requesting*

TAKEAWAY MESSAGES

- Know what you are trying to measure
- Be aware of the different sources of noise
- Simple formulas can predict the expected significance of large signals in white noise
- Simple formulas will miss the mark for small signals in the presence of stellar activity
- Simulating data that roughly represents reality is not too difficult (and it's kinda fun)
- Treat simulated data exactly as you would treat the science data
- Be greedy with your requests

- Beware of overly optimistic simulations. You can simulate anything you want, but that won't change the real data. Try to be realistic.

*THANKS FOR
YOUR ATTENTION*

...and have fun simulating RV data

